

Global Smoothness for the Three-Dimensional Incompressible Navier–Stokes Equations

Thomas Birnie

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Theorem. *For every $\nu > 0$ and every $u_0 \in \mathcal{S}(\mathbb{R}^3)$, there exists a pressure-normalized pair*

$$(u, p) \in C^\infty([0, \infty) \times \mathbb{R}^3; \mathbb{R}^3 \times \mathbb{R})$$

satisfying

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = u_0, \quad \lim_{|x| \rightarrow \infty} p(t, x) = 0,$$

and

$$\frac{1}{2} \|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 \, ds = \frac{1}{2} \|u_0\|_{L^2}^2 \quad (t \geq 0).$$

This pair is unique among pressure-normalized solutions whose velocity belongs to $C([0, T]; H_\sigma^3(\mathbb{R}^3))$ for every finite $T > 0$.

The equations, datum class, energy identity, and smoothness conclusion above are the whole-space formulation of the Navier–Stokes problem [1].

1 Analytic framework

Definition 1.1. For $f \in \mathcal{S}(\mathbb{R}^3)$, set

$$\widehat{f}(\xi) := (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-ix \cdot \xi} f(x) \, dx, \quad \langle \xi \rangle := (1 + |\xi|^2)^{1/2}.$$

The same integral defines $\widehat{f}$ for $f \in L^1(\mathbb{R}^3)$. The Fourier transform on $\mathcal{S}'(\mathbb{R}^3)$ is its extension by duality. For $s \in \mathbb{R}$, define

$$H^s(\mathbb{R}^3) := \left\{ f \in \mathcal{S}'(\mathbb{R}^3) : \langle \xi \rangle^s \widehat{f} \in L^2(\mathbb{R}^3) \right\},$$

and define the finite homogeneous-norm domain by

$$\dot{H}^s(\mathbb{R}^3) := \left\{ f \in \mathcal{S}'(\mathbb{R}^3) : \begin{array}{l} \widehat{f} \text{ is represented by a measurable function on } \mathbb{R}^3, \\ |\xi|^s \widehat{f} \in L^2(\mathbb{R}^3) \end{array} \right\}.$$

On these domains set

$$\|f\|_{H^s}^2 := \int_{\mathbb{R}^3} \langle \xi \rangle^{2s} \left| \widehat{f}(\xi) \right|^2 \, d\xi, \quad \|f\|_{\dot{H}^s}^2 := \int_{\mathbb{R}^3} |\xi|^{2s} \left| \widehat{f}(\xi) \right|^2 \, d\xi.$$

Set $A := (1 - \Delta)^{1/2}$, $\Lambda := (-\Delta)^{1/2}$,

$$H_\sigma^s := \{v \in H^s(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}, \quad H^\infty := \bigcap_{m \in \mathbb{N}_0} H^m.$$

$$\mathcal{S}_\sigma(\mathbb{R}^3) := \{v \in \mathcal{S}(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}.$$

For a vector or tensor $F = (F_I)_I$, all Sobolev norms are componentwise:

$$\|F\|_{H^s}^2 := \sum_I \|F_I\|_{H^s}^2, \quad \|F\|_{\dot{H}^s}^2 := \sum_I \|F_I\|_{\dot{H}^s}^2.$$

Repeated Latin indices are summed from 1 to 3.

Lemma 1.2 (Sobolev nesting and embedding). *If $s_2 \geq s_1$, then*

$$\|f\|_{H^{s_1}} \leq \|f\|_{H^{s_2}}.$$

If $k \in \mathbb{N}_0$ and $s > k + 3/2$, then

$$\max_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^\infty} \leq C_{s,k}^{\text{emb}} \|f\|_{H^s},$$

where

$$C_{s,k}^{\text{emb}} := (2\pi)^{-3/2} \max_{0 \leq j \leq k} \left(\int_{\mathbb{R}^3} |\xi|^{2j} \langle \xi \rangle^{-2s} d\xi \right)^{1/2} < \infty.$$

In particular,

$$H^\infty(\mathbb{R}^3) \hookrightarrow C_b^\infty(\mathbb{R}^3).$$

Proof. The first assertion follows from $\langle \xi \rangle^{2s_1} \leq \langle \xi \rangle^{2s_2}$. For $|\alpha| \leq k$, Fourier inversion and Cauchy-Schwarz give

$$|\partial^\alpha f(x)| \leq (2\pi)^{-3/2} \left(\int_{\mathbb{R}^3} |\xi|^{2|\alpha|} \langle \xi \rangle^{-2s} d\xi \right)^{1/2} \|f\|_{H^s}.$$

The integral is finite exactly when $s > |\alpha| + 3/2$. Approximation by Schwartz functions and dominated convergence give continuous representatives. For each k , choose $s = k + 2$. $\square$

Lemma 1.3 (Explicit three-dimensional homogeneous Sobolev constants). *For $3 \leq p \leq 6$, set*

$$s(p) := 3 \left(\frac{1}{2} - \frac{1}{p} \right)$$

and

$$S(p) := 2^{-s(p)} \pi^{-s(p)/2} \left(\frac{\Gamma((3 - 2s(p))/2)}{\Gamma((3 + 2s(p))/2)} \right)^{1/2} \left(\frac{\Gamma(3)}{\Gamma(3/2)} \right)^{s(p)/3}. \quad (1)$$

Then every scalar, vector, or finite tensor field F in $\dot{H}^{s(p)}(\mathbb{R}^3)$ satisfies

$$\|F\|_{L^p(\mathbb{R}^3)} \leq S(p) \|F\|_{\dot{H}^{s(p)}(\mathbb{R}^3)}. \quad (2)$$

In particular,

$$S(3) = \left[\frac{1}{2\sqrt{\pi}} \left(\frac{4}{\sqrt{\pi}} \right)^{1/3} \right]^{1/2}, \quad S(6) = \left[\frac{1}{3\pi} \left(\frac{4}{\sqrt{\pi}} \right)^{2/3} \right]^{1/2}.$$

Equivalently,

$$S(3) = 2^{-1/6} \pi^{-1/3}, \quad S(6) = \frac{2^{2/3}}{\sqrt{3} \pi^{2/3}}, \quad S(6)S(3) = \frac{1}{\pi} \sqrt{\frac{2}{3}}.$$

Proof. For scalar fields, (2) is the sharp three-dimensional homogeneous Sobolev inequality obtained from the sharp Hardy–Littlewood–Sobolev inequality [5]. The sharp Hardy–Littlewood–Sobolev constant, written for the unitary Fourier transform fixed above and the multiplier $\widehat{\Lambda^s f}(\xi) = |\xi|^s \widehat{f}(\xi)$, is

$$2^{-2s} \pi^{-s} \frac{\Gamma((3-2s)/2)}{\Gamma((3+2s)/2)} \left(\frac{\Gamma(3)}{\Gamma(3/2)} \right)^{2s/3}.$$

Taking its square root and putting $p = 6/(3-2s)$ gives (1)–(2). For a finite component array $F = (F_I)_I$, Minkowski’s inequality in $L^{p/2}$ gives

$$\|F\|_p^2 = \left\| \sum_I |F_I|^2 \right\|_{p/2} \leq \sum_I \|F_I\|_p^2 \leq S(p)^2 \sum_I \|F_I\|_{H^{s(p)}}^2.$$

The two endpoint formulas follow by evaluating the gamma factors at $s = 1/2$ and $s = 1$. $\square$

Lemma 1.4 (Sobolev multiplication). *Set*

$$H_4 := (2\pi)^{-3/4} \left(\int_{\mathbb{R}^3} \langle \xi \rangle^{-4} d\xi \right)^{1/4}, \quad C_0^{\text{alg}} := H_4^2.$$

For $m \geq 1$, define

$$\begin{aligned} E_m &:= \max_{0 \leq j \leq \lfloor m/2 \rfloor} C_{j+2,j}^{\text{emb}}, & C_m^{\text{mix}} &:= 13^{m/2} E_m, \\ C_m^{\text{F}} &:= 2^m (2\pi)^{-3/2} \left(\int_{\mathbb{R}^3} \langle \xi \rangle^{-2m} d\xi \right)^{1/2} & (m \geq 2), \end{aligned}$$

and set

$$C_1^{\text{alg}} := C_1^{\text{mix}}, \quad C_m^{\text{alg}} := \max\{C_m^{\text{mix}}, C_m^{\text{F}}\} \quad (m \geq 2).$$

All these constants are finite and explicit. For every $m \in \mathbb{N}_0$,

$$\|fg\|_{H^m} \leq C_m^{\text{alg}} \|f\|_{H^{m+1}} \|g\|_{H^{m+1}}.$$

For $m \geq 2$,

$$\|fg\|_{H^m} \leq C_m^{\text{alg}} \|f\|_{H^m} \|g\|_{H^m}.$$

Proof. Hausdorff–Young and Hölder give

$$\|f\|_4 \leq (2\pi)^{-3/4} \left\| \widehat{f} \right\|_{4/3} \leq H_4 \|f\|_{H^1}.$$

Thus $\|fg\|_2 \leq C_0^{\text{alg}} \|f\|_{H^1} \|g\|_{H^1}$.

Let $m \geq 1$. The exact integer-order identity

$$\|h\|_{H^m}^2 = \sum_{|\alpha| \leq m} \frac{m!}{(m-|\alpha|)! \alpha!} \|\partial^\alpha h\|_2^2$$

follows by expanding $(1 + |\xi|^2)^m$. In each Leibniz term for $\partial^\alpha(fg)$, place the factor carrying at most $\lfloor |\alpha|/2 \rfloor$ derivatives in L^∞ . The definition of E_m , Sobolev nesting, and $\sum_{\beta \leq \alpha} \binom{\alpha}{\beta} = 2^{|\alpha|}$ give

$$\|\partial^\alpha(fg)\|_2 \leq E_m 2^{|\alpha|} \|f\|_{H^{m+1}} \|g\|_{H^{m+1}}.$$

The multinomial identity

$$\sum_{|\alpha| \leq m} \frac{m!}{(m-|\alpha|)! \alpha!} 4^{|\alpha|} = 13^m$$

proves the first estimate with C_m^{mix} .

For $m \geq 2$, the inequalities

$$\langle \eta + \zeta \rangle^m \leq 2^{m-1} (\langle \eta \rangle^m + \langle \zeta \rangle^m), \quad \|\widehat{f}\|_1 \leq \left(\int_{\mathbb{R}^3} \langle \xi \rangle^{-2m} d\xi \right)^{1/2} \|f\|_{H^m},$$

combined with $\widehat{fg} = (2\pi)^{-3/2} \widehat{f} * \widehat{g}$ and Young's convolution inequality give

$$\|fg\|_{H^m} \leq C_m^F \|f\|_{H^m} \|g\|_{H^m}.$$

The chosen maximum proves both asserted estimates with the same constant. $\square$

Lemma 1.5 (Riesz and Leray multipliers). *Let*

$$r_j(\xi) := \begin{cases} i\xi_j / |\xi|, & \xi \neq 0, \\ 0, & \xi = 0, \end{cases} \quad P(\xi) := \begin{cases} I - \frac{\xi \otimes \xi}{|\xi|^2}, & \xi \neq 0, \\ I, & \xi = 0, \end{cases}$$

and

$$\widehat{R_j f}(\xi) = r_j(\xi) \widehat{f}(\xi), \quad \widehat{\mathbb{P}f}(\xi) = P(\xi) \widehat{f}(\xi).$$

For all $s \in \mathbb{R}$,

$$\|R_j f\|_{H^s} \leq \|f\|_{H^s}, \quad \|\mathbb{P}f\|_{H^s} \leq \|f\|_{H^s}.$$

If $m \in \mathbb{N}_0$ and $v \in H_\sigma^{m+1}$, then

$$q := \sum_{i,j=1}^3 R_i R_j (v_i v_j) \in H^m, \quad -\Delta q = \sum_{i,j=1}^3 \partial_i \partial_j (v_i v_j),$$

and

$$\mathbb{P} \nabla \cdot (v \otimes v) = (v \cdot \nabla) v + \nabla q.$$

Here

$$(v \otimes v)_{ij} := v_i v_j, \quad (\nabla \cdot (v \otimes v))_i := \sum_{j=1}^3 \partial_j (v_i v_j), \quad ((v \cdot \nabla) v)_i := \sum_{j=1}^3 v_j \partial_j v_i.$$

Proof. The multiplier symbols have operator norm at most one. Plancherel proves the first two estimates. Lemma 1.4 gives $v_i v_j \in H^m$. The remaining identities follow after multiplication by $|\xi|^2$ and by the symbol of $\mathbb{P}$. $\square$

Lemma 1.6 (Mixed Leibniz identities). *For $N \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$\partial_t^N \partial_x^\alpha (fg) = \sum_{a=0}^N \sum_{\beta \leq \alpha} \binom{N}{a} \binom{\alpha}{\beta} (\partial_t^a \partial_x^\beta f) (\partial_t^{N-a} \partial_x^{\alpha-\beta} g).$$

Proof. Apply the one-variable Leibniz identity successively in t, x_1, x_2, x_3 . $\square$

Lemma 1.7 (Adjacent Hilbert-scale trace). *Let $0 < T < \infty$, $m \in \mathbb{R}$, and*

$$v \in L^2(0, T; H^{m+1}), \quad \partial_t v \in L^2(0, T; H^{m-1}).$$

Then v has a unique representative in $C([0, T]; H^m)$, and

$$\|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2 = 2 \operatorname{Re} \int_r^t \langle A^{m-1} \partial_t v(s), A^{m+1} v(s) \rangle_{L^2} ds, \quad (3)$$

$$\left| \|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2 \right| \leq 2 \|\partial_t v\|_{L^2(r, t; H^{m-1})} \|v\|_{L^2(r, t; H^{m+1})}, \quad (4)$$

$$\sup_{0 \leq t \leq T} \|v(t)\|_{H^m}^2 \leq T^{-1} \|v\|_{L^2(0, T; H^m)}^2 + 2 \|\partial_t v\|_{L^2(0, T; H^{m-1})} \|v\|_{L^2(0, T; H^{m+1})}. \quad (5)$$

Proof. Let S_R denote Fourier restriction to $\{|\xi| \leq R\}$. For $v_R = S_R v$,

$$\frac{d}{dt} \|v_R(t)\|_{H^m}^2 = 2 \operatorname{Re} \langle A^{m-1} \partial_t v_R(t), A^{m+1} v_R(t) \rangle_{L^2}.$$

Integration and Cauchy–Schwarz prove (3)–(4) for v_R . Applied to $S_R v - S_S v$, the same estimate shows that $(v_R)_R$ is Cauchy in $C([0, T]; H^m)$; choose one Lebesgue time for the initial Cauchy term. Its limit equals v in $L^2(0, T; H^m)$. Passage to the limit proves the first two formulas. Choose $r \in (0, T)$ with $\|v(r)\|_{H^m}^2 \leq T^{-1} \|v\|_{L^2 H^m}^2$ and apply (4). $\square$

Lemma 1.8 (Frequency tail). *If $s > \sigma$, $K \geq 1$, and $f \in \dot{H}^s(\mathbb{R}^3)$, then*

$$\int_{|\xi| > K} |\xi|^{2\sigma} |\widehat{f}(\xi)|^2 d\xi \leq K^{-2(s-\sigma)} \|f\|_{\dot{H}^s}^2.$$

Proof. On $\{|\xi| > K\}$, $|\xi|^{2\sigma} \leq K^{-2(s-\sigma)} |\xi|^{2s}$. $\square$

Lemma 1.9 (Integer transport commutator). *For every integer $m \geq 3$, there is $C_m^{\text{com}} < \infty$ such that every $v \in H_\sigma^{m+1}(\mathbb{R}^3)$ satisfies*

$$|\operatorname{Re} \langle A^m \mathbb{P} \nabla \cdot (v \otimes v), A^m v \rangle_{L^2}| \leq C_m^{\text{com}} \|\nabla v\|_{L^\infty} \|v\|_{H^m}^2. \quad (6)$$

Proof. For $|\alpha| \leq m$, set

$$\alpha! := \alpha_1! \alpha_2! \alpha_3!, \quad c_{m, \alpha} := \frac{m!}{(m - |\alpha|)! \alpha!}.$$

The multiplier $A^{2m} = (1 - \Delta)^m$ and the multinomial identity give the exact pairing formula

$$\langle A^m f, A^m g \rangle_{L^2} = \sum_{|\alpha| \leq m} c_{m, \alpha} \langle \partial^\alpha f, \partial^\alpha g \rangle_{L^2}. \quad (7)$$

The Leray projection is self-adjoint, commutes with A^m , and fixes $A^m v$. Since $\nabla \cdot (v \otimes v) = (v \cdot \nabla) v$, it follows that

$$\langle A^m \mathbb{P} \nabla \cdot (v \otimes v), A^m v \rangle_{L^2} = \sum_{|\alpha| \leq m} c_{m, \alpha} \langle \partial^\alpha ((v \cdot \nabla) v), \partial^\alpha v \rangle_{L^2}. \quad (8)$$

For each α ,

$$\partial^\alpha ((v \cdot \nabla) v) = (v \cdot \nabla) \partial^\alpha v + \sum_{0 < \beta \leq \alpha} \binom{\alpha}{\beta} (\partial^\beta v \cdot \nabla) \partial^{\alpha-\beta} v.$$

The first term has zero real pairing with $\partial^\alpha v$, because

$$\operatorname{Re} \langle (v \cdot \nabla) \partial^\alpha v, \partial^\alpha v \rangle_{L^2} = \frac{1}{2} \int_{\mathbb{R}^3} v \cdot \nabla |\partial^\alpha v|^2 \, dx = 0.$$

For a remaining term, put $\ell = |\alpha|$ and $k = |\beta|$. When $\ell \geq 2$, apply the Gagliardo–Nirenberg inequalities to $f = \nabla v$, with $M = \ell - 1$, $j = k - 1$, and the endpoint conventions at $j = 0, M$:

$$\|D^j f\|_{L^{2M/j}} \leq C_M \|f\|_{L^\infty}^{1-j/M} \|D^M f\|_{L^2}^{j/M}.$$

Here $L^{2M/0}$ means L^∞ . Hölder's inequality with the exponents belonging to j and $M - j$, whose reciprocal sum is $1/2$, yields

$$\left\| (\partial^\beta v \cdot \nabla) \partial^{\alpha-\beta} v \right\|_{L^2} \leq C_m \|\nabla v\|_{L^\infty} \|v\|_{H^\ell}.$$

The same estimate for $\ell = 1$ is the direct $L^\infty \cdot L^2$ bound. Multiplication by $\|\partial^\alpha v\|_2$, summation over β, α , and (8) prove (6). Approximation by divergence-free Schwartz fields justifies the integrations by parts for $v \in H_\sigma^{m+1}$. $\square$

Theorem 1.10 (Quantitative local restart). *Set $C_q := \sqrt{3} C_3^{\text{alg}}$. For $\nu > 0$, set*

$$L_\nu := 4(1 + \nu + \nu^{-1}) \exp\left(\frac{\nu + \nu^{-1}}{2}\right), \quad c_\nu := (8C_q L_\nu^2)^{-2}.$$

Then every $a \in H_\sigma^3(\mathbb{R}^3)$ generates a unique solution

$$w \in C([0, \delta]; H_\sigma^3) \cap L^2(0, \delta; H^4), \quad w_t \in L^2(0, \delta; H^2),$$

for every

$$0 < \delta \leq \delta_\nu(a) := \min\{1, c_\nu(1 + \|a\|_{H^3})^{-2}\}.$$

Its normalized pressure is $q = R_i R_j (w_i w_j)$. If $a \in H_\sigma^\infty$, then

$$\partial_t^n w, \partial_t^n q \in C([0, \delta]; H^m) \quad (n, m \in \mathbb{N}_0).$$

Proof. Let $0 < \delta \leq 1$, let

$$z_t - \nu \Delta z = F, \quad z(0) = a, \quad a \in H_\sigma^3, \quad F \in L^2(0, \delta; H_\sigma^2),$$

and set

$$A_0 := \|a\|_{H^3}, \quad G := \|F\|_{L^2(0, \delta; H^2)}.$$

The H^3 energy identity is

$$\frac{1}{2} \frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 = \operatorname{Re} \langle A^2 F, A^4 z \rangle_{L^2}.$$

Since $\|z\|_{H^4}^2 = \|z\|_{H^3}^2 + \|\nabla z\|_{H^3}^2$, Young's inequality gives

$$\frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 \leq \nu \|z\|_{H^3}^2 + \nu^{-1} \|F\|_{H^2}^2.$$

Consequently,

$$\|z\|_{C([0, \delta]; H^3)} \leq e^{\nu/2} (A_0 + \nu^{-1/2} G). \quad (9)$$

Pairing $A^2 z_t - \nu \Delta A^2 z = A^2 F$ with $-\Delta A^2 z$ gives

$$\frac{1}{2} \frac{d}{dt} \|\nabla A^2 z\|_2^2 + \nu \|\Delta A^2 z\|_2^2 = \operatorname{Re} \langle A^2 F, -\Delta A^2 z \rangle_{L^2}.$$

Integration and Young's inequality yield

$$\|\Delta z\|_{L^2(0,\delta;H^2)} \leq \nu^{-1/2} A_0 + \nu^{-1} G. \quad (10)$$

The pointwise multiplier inequality

$$\langle \xi \rangle^8 \leq 2(\langle \xi \rangle^6 + |\xi|^4 \langle \xi \rangle^4)$$

and $\delta \leq 1$ imply

$$\|z\|_{L^2(0,\delta;H^4)} \leq \sqrt{2} \left(\|z\|_{C([0,\delta];H^3)} + \|\Delta z\|_{L^2(0,\delta;H^2)} \right). \quad (11)$$

The equation and (10) give

$$\|z_t\|_{L^2(0,\delta;H^2)} \leq \nu^{1/2} A_0 + 2G. \quad (12)$$

Equations (9)–(12) give

$$\begin{aligned} & \|z\|_{CH^3} + \|z\|_{L^2 H^4} + \|z_t\|_{L^2 H^2} \\ & \leq e^{\nu/2} (A_0 + \nu^{-1/2} G) + \sqrt{2} \left(e^{\nu/2} (A_0 + \nu^{-1/2} G) + \nu^{-1/2} A_0 + \nu^{-1} G \right) + \nu^{1/2} A_0 + 2G \\ & \leq L_\nu (A_0 + G). \end{aligned}$$

The last inequality follows from

$$\nu^{1/2} \leq \frac{1+\nu}{2}, \quad \nu^{-1/2} \leq \frac{1+\nu^{-1}}{2}, \quad e^{\nu/2} \leq e^{(\nu+\nu^{-1})/2}.$$

Fourier truncation and passage to the limit justify all four linear identities for the stated data class.

For $v, z \in H_\sigma^3$, Lemmas 1.4 and 1.5 give

$$\|\mathbb{P} \nabla \cdot (v \otimes z)\|_{H^2} \leq C_q \|v\|_{H^3} \|z\|_{H^3}, \quad (13)$$

$$\|\mathbb{P} \nabla \cdot (v \otimes v - z \otimes z)\|_{H^2} \leq C_q (\|v\|_{H^3} + \|z\|_{H^3}) \|v - z\|_{H^3}. \quad (14)$$

Indeed,

$$\|\nabla \cdot (v \otimes z)\|_{H^2}^2 \leq 3 \sum_{i,j=1}^3 \|v_i z_j\|_{H^3}^2 \leq 3(C_3^{\text{alg}})^2 \|v\|_{H^3}^2 \|z\|_{H^3}^2.$$

Define

$$X_\delta := \{v \in C([0, \delta]; H_\sigma^3) \cap L^2(0, \delta; H_\sigma^4) : v_t \in L^2(0, \delta; H_\sigma^2)\},$$

$$\|v\|_{X_\delta} := \|v\|_{CH^3} + \|v\|_{L^2 H^4} + \|v_t\|_{L^2 H^2}, \quad X_\delta(a) := \{v \in X_\delta : v(0) = a\}.$$

X_δ is a Banach space with this norm, and $X_\delta(a)$ is closed. For $A_0 > 0$, set $R := 2L_\nu A_0$ and

$$\mathbb{B}_R := \{v \in X_\delta(a) : \|v\|_{X_\delta} \leq R\}.$$

The set $\mathbb{B}_R$ is nonempty: the zero-forcing heat trajectory $v^{\text{lin}}(t) = e^{\nu t \Delta} a$ belongs to $X_\delta(a)$, and the linear estimate with $F = 0$ gives $\|v^{\text{lin}}\|_{X_\delta} \leq L_\nu A_0 < R$. For $v \in \mathbb{B}_R$, define

$$\Phi(v)(t) := e^{\nu t \Delta} a - \int_0^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (v \otimes v)(s) \, ds.$$

If $\delta \leq \delta_\nu(a)$, then

$$\delta^{1/2} \leq \frac{1}{8C_q L_\nu^2 (1 + A_0)}.$$

The linear estimate and (13) imply

$$\begin{aligned} \|\Phi(v)\|_{X_\delta} &\leq L_\nu (A_0 + C_q \delta^{1/2} R^2) \\ &\leq L_\nu A_0 + \frac{L_\nu A_0^2}{2(1 + A_0)} < R. \end{aligned}$$

For $v, z \in \mathbb{B}_R$,

$$\begin{aligned} \|\Phi(v) - \Phi(z)\|_{X_\delta} &\leq L_\nu C_q \delta^{1/2} (\|v\|_{CH^3} + \|z\|_{CH^3}) \|v - z\|_{CH^3} \\ &\leq \frac{A_0}{2(1 + A_0)} \|v - z\|_{X_\delta} \leq \frac{1}{2} \|v - z\|_{X_\delta}. \end{aligned}$$

Thus Φ has a unique fixed point in $\mathbb{B}_R$. If $A_0 = 0$, the zero function is the fixed point. For any two mild solutions $v, z \in C([0, \delta]; H_\sigma^3)$, set

$$M := \|v\|_{CH^3} + \|z\|_{CH^3}.$$

On every subinterval of length $h \leq 1$ on which the two solutions have the same initial value, the zero-data linear estimate gives

$$\|v - z\|_{CH^3} \leq L_\nu C_q h^{1/2} M \|v - z\|_{CH^3}.$$

A finite partition with $L_\nu C_q h^{1/2} M < 1$ proves uniqueness in the full class $C([0, \delta]; H_\sigma^3)$. For the fixed point, (13) places the forcing in $L^2(0, \delta; H^2)$; the linear estimates give

$$w \in L^2(0, \delta; H^4), \quad w_t \in L^2(0, \delta; H^2).$$

Lemma 1.5 gives the normalized pressure $q = R_i R_j (w_i w_j)$.

For $m \geq 3$, Lemma 1.9 gives

$$\frac{1}{2} \frac{d}{dt} \|w\|_{H^m}^2 + \nu \|\nabla w\|_{H^m}^2 \leq C_m^{\text{com}} \|\nabla w\|_{L^\infty} \|w\|_{H^m}^2.$$

Since $w \in CH^3 \hookrightarrow CW^{1,\infty}$, Gronwall's inequality and Fourier-cutoff approximation give, on the same interval,

$$a \in H_\sigma^m \implies w \in CH_\sigma^m \cap L^2 H_\sigma^{m+1}, \quad w_t \in L^2 H_\sigma^{m-1}.$$

Assume now that $a \in H_\sigma^\infty$. Set $W_0 = w$ and define recursively

$$\begin{aligned} Q_n &= R_i R_j \sum_{a=0}^n \binom{n}{a} W_{a,i} W_{n-a,j}, \\ W_{n+1} &= \nu \Delta W_n - \sum_{a=0}^n \binom{n}{a} (W_a \cdot \nabla) W_{n-a} - \nabla Q_n. \end{aligned}$$

If $W_0, \dots, W_n \in CH^{m+2}$, then Lemmas 1.4 and 1.5 give

$$Q_n \in CH^{m+1}, \quad W_{n+1} \in CH^m.$$

Since m is arbitrary, induction gives $W_n, Q_n \in CH^m$ for every $n, m \in \mathbb{N}_0$. The equation first gives $W_0 \in C^1 H^m$ and $\partial_t W_0 = W_1$. If $W_a = \partial_t^a w$ for $0 \leq a \leq n$, distributional differentiation of $q = R_i R_j (w_i w_j)$ and Lemma 1.6 give $\partial_t^n q = Q_n$. Differentiation of the momentum equation then gives $\partial_t W_n = W_{n+1}$. Induction gives

$$W_n \in C^1 H^m, \quad \partial_t W_n = W_{n+1}, \quad W_n = \partial_t^n w, \quad Q_n = \partial_t^n q.$$

□

2 The pressure-complete setting and mixed jet

Proposition 2.1 (Pressure normalization). *Let u, p be smooth on $I \times \mathbb{R}^3$, let $\nabla \cdot u = 0$, and suppose*

$$u(t) \in H_\sigma^1(\mathbb{R}^3), \quad p(t) \in H^0(\mathbb{R}^3) \quad (t \in I),$$

and

$$u_t + (u \cdot \nabla)u + \nabla p = \nu \Delta u. \quad (15)$$

Then

$$-\Delta p = \partial_i \partial_j (u_i u_j), \quad (16)$$

$$p = R_i R_j (u_i u_j), \quad (17)$$

$$(u \cdot \nabla)u + \nabla p = \mathbb{P} \nabla \cdot (u \otimes u). \quad (18)$$

The condition $p(t) \in H^0(\mathbb{R}^3)$ fixes the pressure uniquely.

Proof. Incompressibility gives

$$(u \cdot \nabla)u_i = \partial_j (u_j u_i).$$

Taking the divergence of (15) cancels $\nabla \cdot u_t$ and $\nabla \cdot \Delta u$, which yields (16). Lemma 1.5, applied with $m = 0$, gives $q := R_i R_j (u_i u_j) \in H^0$, the same Poisson identity, and

$$(u \cdot \nabla)u + \nabla q = \mathbb{P} \nabla \cdot (u \otimes u).$$

If $p_1, p_2 \in H^0$ satisfy (16), then $h = p_1 - p_2$ satisfies $|\xi|^2 \widehat{h} = 0$. Since $\widehat{h} \in L^2$ and is supported at $\{0\}$, it vanishes. Hence $p = q$, which proves (17) and (18). $\square$

Definition 2.2 (Maximal pressure-complete classical history). Fix $\nu > 0$ and $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$. A pressure-complete classical history generated by (u_0, ν) is a triple

$$(T_*, u, p), \quad 0 < T_* \leq \infty,$$

such that, for every $0 < T < T_*$ and $n, m \in \mathbb{N}_0$,

$$\partial_t^n u \in C([0, T]; H_\sigma^m(\mathbb{R}^3)),$$

$$\partial_t^n p \in C([0, T]; H^m(\mathbb{R}^3)),$$

and

$$u_t + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad (19)$$

$$\nabla \cdot u = 0, \quad (20)$$

$$u(0) = u_0, \quad (21)$$

$$p(t) = R_i R_j (u_i(t) u_j(t)). \quad (22)$$

The history is maximal if no pressure-complete classical history $(\widetilde{T}, \widetilde{u}, \widetilde{p})$ generated by (u_0, ν) satisfies

$$\widetilde{T} > T_*, \quad (\widetilde{u}, \widetilde{p})|_{[0, T_*]} = (u, p).$$

Definition 2.3 (Temporal and mixed velocity–pressure jets). Let (T_*, u, p) be a maximal pressure-complete classical history. For $n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$, define

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p, \quad J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p.$$

For $\beta \leq \alpha$, define

$$\binom{\alpha}{\beta} := \prod_{\ell=1}^3 \binom{\alpha_\ell}{\beta_\ell}, \quad |\alpha| := \alpha_1 + \alpha_2 + \alpha_3.$$

Proposition 2.4 (Pressure-complete binomial recurrences). *For every $n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}, \quad (23)$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n, \quad (24)$$

$$\Pi_{n,\alpha} = R_i R_j \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} J_{a,\beta,i} J_{n-a,\alpha-\beta,j}, \quad (25)$$

$$J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} + \nabla \Pi_{n,\alpha} = \nu \Delta J_{n,\alpha}, \quad (26)$$

$$\nabla \cdot J_{n,\alpha} = 0. \quad (27)$$

Proof. Apply ∂_t^n to (22) and use Lemma 1.6 with $\alpha = 0$ to obtain (23). Apply ∂_t^n to (19) to obtain (24). Apply ∂_x^α to both identities, use Lemma 1.6, and commute the derivatives with $R_i R_j$ to obtain (25) and (26). Differentiation of (20) gives (27). $\square$

Proposition 2.5 (Mixed-index transport matrix and pressure completion). *Let*

$$\Gamma := \mathbb{N}_0 \times \mathbb{N}_0^3, \quad \kappa = (n, \alpha), \quad \lambda = (a, \beta),$$

and write

$$\lambda \preceq \kappa \iff a \leq n \text{ and } \beta \leq \alpha, \quad \kappa - \lambda := (n - a, \alpha - \beta), \quad \binom{\kappa}{\lambda} := \binom{n}{a} \binom{\alpha}{\beta}.$$

Set

$$J_\kappa := J_{n,\alpha}, \quad \Pi_\kappa := \Pi_{n,\alpha}, \quad \mathbf{J} := (J_\lambda)_{\lambda \in \Gamma}.$$

The block operator

$$\mathbb{L}(J)_{\kappa,\lambda} := \begin{cases} \binom{\kappa}{\lambda} \mathbb{P}[(J_{\kappa-\lambda} \cdot \nabla) \cdot], & \lambda \preceq \kappa, \\ 0, & \lambda \not\preceq \kappa, \end{cases} \quad (28)$$

has the finite block-row action

$$[\mathbb{L}(J)\mathbf{J}]_\kappa = \sum_{\lambda \preceq \kappa} \binom{\kappa}{\lambda} \mathbb{P}[(J_\lambda \cdot \nabla) J_{\kappa-\lambda}]. \quad (29)$$

Consequently the complete mixed row is the pair

$$J_{n+1,\alpha} + [\mathbb{L}(J)\mathbf{J}]_\kappa = \nu \Delta J_\kappa, \quad (30)$$

$$\Pi_\kappa = R_i R_j \sum_{\lambda \preceq \kappa} \binom{\kappa}{\lambda} J_{\lambda,i} J_{\kappa-\lambda,j}, \quad (31)$$

while the unprojected velocity identity retains $\nabla \Pi_\kappa$ exactly as in (26).

Order Γ by increasing $n + |\alpha|$, with any fixed ordering among indices of equal depth. Then $\mathbb{L}(J)$ is block lower triangular. For

$$\kappa! := n! \alpha!, \quad \alpha! := \alpha_1! \alpha_2! \alpha_3!, \quad \hat{J}_\kappa := \frac{J_\kappa}{\kappa!}, \quad \hat{\Pi}_\kappa := \frac{\Pi_\kappa}{\kappa!},$$

let D be the diagonal matrix with $D_{\kappa,\kappa} = \kappa!$. On every finite downward-closed truncation of Γ ,

$$D^{-1} \mathbb{L}(D \hat{J}) D = \hat{\mathbb{L}}(\hat{J}), \quad \hat{\mathbb{L}}(\hat{J})_{\kappa,\lambda} := \begin{cases} \mathbb{P}[(\hat{J}_{\kappa-\lambda} \cdot \nabla) \cdot], & \lambda \preceq \kappa, \\ 0, & \lambda \not\preceq \kappa. \end{cases} \quad (32)$$

Thus the factorial-normalized pressure-complete recurrence is

$$(n+1) \hat{J}_{n+1,\alpha} + \sum_{\lambda \preceq \kappa} (\hat{J}_\lambda \cdot \nabla) \hat{J}_{\kappa-\lambda} + \nabla \hat{\Pi}_\kappa = \nu \Delta \hat{J}_\kappa, \quad (33)$$

$$\hat{\Pi}_\kappa = R_i R_j \sum_{\lambda \preceq \kappa} \hat{J}_{\lambda,i} \hat{J}_{\kappa-\lambda,j}. \quad (34)$$

The normalized transport blocks depend on the difference $\kappa - \lambda$; hence (32) is the block lower-triangular convolution form of the mixed transport array.

Proof. For $\lambda \preceq \kappa$, change the summation index from λ to $\kappa - \lambda$ and use $\binom{\kappa}{\kappa-\lambda} = \binom{\kappa}{\lambda}$. This gives (29). Apply $\mathbb{P}$ to (26). The identities

$$\mathbb{P} J_{n+1,\alpha} = J_{n+1,\alpha}, \quad \mathbb{P} \Delta J_\kappa = \Delta J_\kappa, \quad \mathbb{P} \nabla \Pi_\kappa = 0$$

follow from (27) and the Fourier symbols. They prove (30); equation (31) is (25) in mixed-index notation. The pressure formula also shows that projection of the velocity row loses no pressure coordinate.

If $\lambda \preceq \kappa$, then $a + |\beta| \leq n + |\alpha|$, which proves lower triangularity in the stated ordering. Finally,

$$\binom{\kappa}{\lambda} \lambda! (\kappa - \lambda)! = \kappa!.$$

Dividing the mixed velocity and pressure rows by $\kappa!$ proves (33) and (34). The same factorial cancellation in each operator block proves (32). Every row has $(n+1) \prod_{j=1}^3 (\alpha_j + 1)$ supported blocks, so all displayed row sums are finite. $\square$

Definition 2.6 (Canonical transport and pressure constants). For vector fields f, g , set

$$\mathcal{B}(f, g) := (f \cdot \nabla) g, \quad \mathcal{Q}(f, g) := \sum_{i,j=1}^3 R_i R_j (f_i g_j).$$

For $m \in \mathbb{N}_0$, define

$$C_m^{\text{tr}} := \sup_{\substack{f \in H^{m+1}(\mathbb{R}^3; \mathbb{R}^3) \setminus \{0\} \\ g \in H^{m+2}(\mathbb{R}^3; \mathbb{R}^3) \setminus \{0\}}} \frac{\|\mathcal{B}(f, g)\|_{H^m}}{\|f\|_{H^{m+1}} \|g\|_{H^{m+2}}},$$

$$C_m^{\text{p}} := \sup_{f, g \in H^{m+1}(\mathbb{R}^3; \mathbb{R}^3) \setminus \{0\}} \frac{\|\mathcal{Q}(f, g)\|_{H^m} + \|\nabla \mathcal{Q}(f, g)\|_{H^{m-1}}}{\|f\|_{H^{m+1}} \|g\|_{H^{m+1}}}.$$

Lemma 2.7. *For every $m \in \mathbb{N}_0$,*

$$C_m^{\text{tr}} \leq C_m^{\text{alg}}, \quad C_m^{\text{p}} \leq 2C_m^{\text{alg}}.$$

Proof. Lemma 1.4 and componentwise Cauchy–Schwarz give

$$\|\mathcal{B}(f, g)\|_{H^m} \leq C_m^{\text{alg}} \|f\|_{H^{m+1}} \|g\|_{H^{m+2}}.$$

For pressure, regard $(f_i g_j)_{i,j}$ as one tensor. The contraction $F \mapsto R_i R_j F_{ij}$ has Fourier-multiplier norm one from the tensor H^m norm to scalar H^m . The same product lemma and $\|\nabla h\|_{H^{m-1}} \leq \|h\|_{H^m}$ give

$$\|\mathcal{Q}(f, g)\|_{H^m} + \|\nabla \mathcal{Q}(f, g)\|_{H^{m-1}} \leq 2C_m^{\text{alg}} \|f\|_{H^{m+1}} \|g\|_{H^{m+1}}.$$

□

Lemma 2.8 (Initial pressure-complete jet). *Set*

$$V_n^0 := V_n(0), \quad Q_n^0 := Q_n(0), \quad J_{n,\alpha}^0 := J_{n,\alpha}(0), \quad \Pi_{n,\alpha}^0 := \Pi_{n,\alpha}(0).$$

Then

$$V_0^0 = u_0, \tag{35}$$

$$Q_n^0 = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}^0 V_{n-a,j}^0, \tag{36}$$

$$V_{n+1}^0 = \nu \Delta V_n^0 - \sum_{a=0}^n \binom{n}{a} (V_a^0 \cdot \nabla) V_{n-a}^0 - \nabla Q_n^0. \tag{37}$$

For every $n, m \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,

$$V_n^0, Q_n^0, J_{n,\alpha}^0, \Pi_{n,\alpha}^0 \in H^m(\mathbb{R}^3).$$

Moreover,

$$\|Q_n^0\|_{H^m} \leq C_m^{\text{p}} \sum_{a=0}^n \binom{n}{a} \|V_a^0\|_{H^{m+1}} \|V_{n-a}^0\|_{H^{m+1}}, \tag{38}$$

$$\|V_{n+1}^0\|_{H^m} \leq \nu \|V_n^0\|_{H^{m+2}} + C_m^{\text{tr}} \sum_{a=0}^n \binom{n}{a} \|V_a^0\|_{H^{m+1}} \|V_{n-a}^0\|_{H^{m+2}} + \|Q_n^0\|_{H^{m+1}}. \tag{39}$$

For $m \in \mathbb{N}_0$, set

$$\mathcal{U}_m(u_0)^2 := \sum_{\substack{\gamma \in \mathbb{N}_0^3 \\ |\gamma| \leq m}} \frac{m!}{(m - |\gamma|)! \gamma!} \|\partial^\gamma u_0\|_{L^2}^2, \quad \gamma! := \gamma_1! \gamma_2! \gamma_3!,$$

and define $D_{n,m} = D_{n,m}(u_0, \nu)$ and $E_{n,m} = E_{n,m}(u_0, \nu)$ recursively by

$$D_{0,m} := \mathcal{U}_m(u_0), \tag{40}$$

$$E_{n,m} := C_m^{\text{p}} \sum_{a=0}^n \binom{n}{a} D_{a,m+1} D_{n-a,m+1}, \tag{41}$$

$$D_{n+1,m} := \nu D_{n,m+2} + C_m^{\text{tr}} \sum_{a=0}^n \binom{n}{a} D_{a,m+1} D_{n-a,m+2} + E_{n,m+1}. \quad (42)$$

Then

$$\|V_n^0\|_{H^m} \leq D_{n,m}, \quad \|Q_n^0\|_{H^m} \leq E_{n,m}, \quad (43)$$

$$\|J_{n,\alpha}^0\|_{H^m} \leq D_{n,m+|\alpha|}, \quad \|\Pi_{n,\alpha}^0\|_{H^m} \leq E_{n,m+|\alpha|}. \quad (44)$$

For fixed n, m , $D_{n,m}$ depends only on

$$\nu, \quad \{\|\partial^\gamma u_0\|_{L^2} : |\gamma| \leq m + 2n\},$$

and $E_{n,m}$ depends only on

$$\nu, \quad \{\|\partial^\gamma u_0\|_{L^2} : |\gamma| \leq m + 2n + 1\}.$$

Proof. Equations (35)–(37) are (23) and (24) at $t = 0$. Assume $V_0^0, \dots, V_n^0 \in \bigcap_{m \geq 0} H^m$. Lemmas 1.4 and 1.5 give

$$Q_n^0 \in \bigcap_{m \geq 0} H^m, \quad V_{n+1}^0 \in \bigcap_{m \geq 0} H^m.$$

Induction starts from $u_0 \in \mathcal{S}_\sigma$. Spatial differentiation and (25) give the claims for $J_{n,\alpha}^0$ and $\Pi_{n,\alpha}^0$.

The definition of C_m^{p} applied to (36) proves (38). The definition of C_m^{tr} , the inequality $\|\nabla h\|_{H^m} \leq \|h\|_{H^{m+1}}$, and (37) prove (39).

The multinomial identity

$$(1 + |\xi|^2)^m = \sum_{\substack{\gamma \in \mathbb{N}_0^3 \\ |\gamma| \leq m}} \frac{m!}{(m - |\gamma|)! \gamma!} \xi^{2\gamma}$$

and Plancherel give

$$\mathcal{U}_m(u_0) = \|u_0\|_{H^m}. \quad (45)$$

Equations (38) and (39), followed by induction on n , give (43). For every $\alpha \in \mathbb{N}_0^3$,

$$\|\partial^\alpha f\|_{H^m} \leq \|f\|_{H^{m+|\alpha|}}.$$

Since $J_{n,\alpha}^0 = \partial^\alpha V_n^0$ and $\Pi_{n,\alpha}^0 = \partial^\alpha Q_n^0$, this proves (44). A second induction on n in (41)–(42) gives the stated maximal derivative orders of u_0 . $\square$

Theorem 2.9 (Finite-row pressure estimates). *Let $I = (0, T) \subset (0, T_*)$, let $n, M \in \mathbb{N}_0$, and suppose*

$$V_a \in L^2(I; H^{M+1}(\mathbb{R}^3)) \quad (0 \leq a \leq n).$$

Then

$$\begin{aligned} & \|Q_n\|_{L^1(I; H^M)} + \|\nabla Q_n\|_{L^1(I; H^{M-1})} \\ & \leq C_M^{\text{p}} \sum_{a=0}^n \binom{n}{a} \|V_a\|_{L^2(I; H^{M+1})} \|V_{n-a}\|_{L^2(I; H^{M+1})}. \end{aligned} \quad (46)$$

If

$$\mathcal{V}_{n,M}(I)^2 := \sum_{a=0}^n \|V_a\|_{L^2(I; H^{M+1})}^2,$$

then

$$\|Q_n\|_{L^1(I;H^M)} + \|\nabla Q_n\|_{L^1(I;H^{M-1})} \leq C_M^p 2^n \mathcal{V}_{n,M}(I)^2. \quad (47)$$

For $\alpha \in \mathbb{N}_0^3$, assume

$$J_{a,\beta} \in L^2(I;H^{M+1}(\mathbb{R}^3)) \quad (0 \leq a \leq n, \beta \leq \alpha).$$

Then

$$\begin{aligned} & \|\Pi_{n,\alpha}\|_{L^1(I;H^M)} + \|\nabla \Pi_{n,\alpha}\|_{L^1(I;H^{M-1})} \\ & \leq C_M^p \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} \|J_{a,\beta}\|_{L^2(I;H^{M+1})} \|J_{n-a,\alpha-\beta}\|_{L^2(I;H^{M+1})}. \end{aligned} \quad (48)$$

If

$$\mathcal{J}_{n,\alpha,M}(I)^2 := \sum_{a=0}^n \sum_{\beta \leq \alpha} \|J_{a,\beta}\|_{L^2(I;H^{M+1})}^2,$$

then

$$\|\Pi_{n,\alpha}\|_{L^1(I;H^M)} + \|\nabla \Pi_{n,\alpha}\|_{L^1(I;H^{M-1})} \leq C_M^p 2^{n+|\alpha|} \mathcal{J}_{n,\alpha,M}(I)^2. \quad (49)$$

Proof. Lemmas 1.4 and 1.5 give

$$\|R_i R_j(f_i g_j)\|_{H^M} + \|\nabla R_i R_j(f_i g_j)\|_{H^{M-1}} \leq C_M^p \|f\|_{H^{M+1}} \|g\|_{H^{M+1}}. \quad (50)$$

Equations (23) and (50) give, for almost every $t \in I$,

$$\|Q_n(t)\|_{H^M} + \|\nabla Q_n(t)\|_{H^{M-1}} \leq C_M^p \sum_{a=0}^n \binom{n}{a} \|V_a(t)\|_{H^{M+1}} \|V_{n-a}(t)\|_{H^{M+1}}.$$

Integration and Cauchy–Schwarz prove (46). For nonnegative x_a ,

$$\sum_{a=0}^n \binom{n}{a} x_a x_{n-a} \leq \sum_{a=0}^n \binom{n}{a} x_a^2 \leq 2^n \sum_{a=0}^n x_a^2.$$

This proves (47).

Apply (50) to each summand in (25). Integration and Cauchy–Schwarz give (48). The involution

$$(a, \beta) \mapsto (n-a, \alpha-\beta)$$

preserves the weights $\binom{n}{a} \binom{\alpha}{\beta}$, and

$$\sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} = 2^{n+|\alpha|}.$$

In particular, every individual weight is at most $2^{n+|\alpha|}$. The inequality $2xy \leq x^2 + y^2$ proves (49). $\square$

3 Datum-generated compatible rectangles

Proposition 3.1 (Kinetic-energy identity). *For every $0 \leq t < T_*$,*

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \, ds = \frac{1}{2} \|u_0\|_2^2. \quad (51)$$

Consequently, for $0 < T < T_$,*

$$\|u\|_{L^\infty(0,T;L^2)} \leq \|u_0\|_2, \quad (52)$$

$$\|u\|_{L^2(0,T;H^1)}^2 \leq \left(T + \frac{1}{2\nu}\right) \|u_0\|_2^2. \quad (53)$$

Proof. Take the L^2 pairing of the momentum equation with u . Integration by parts and $\nabla \cdot u = 0$ give

$$\langle (u \cdot \nabla)u, u \rangle_{L^2} = 0, \quad \langle \nabla p, u \rangle_{L^2} = 0, \quad \langle \Delta u, u \rangle_{L^2} = -\|\nabla u\|_2^2.$$

Integration in time proves (51). The two estimates follow by discarding the nonnegative dissipation and adding $\int_0^T \|u(s)\|_2^2 \, ds$. $\square$

Proposition 3.2 (Datum-paid kinetic base block). *Let*

$$\mathcal{B}(u) := \mathbb{P} \nabla \cdot (u \otimes u),$$

and define the explicit three-dimensional constants

$$\begin{aligned} \mathbf{G}_4 &:= \mathbf{S}(6)^{3/4}, \\ \mathbf{L}_{-1} &:= 1, \\ \mathbf{R}_2 &:= 1, \\ C_b &:= \mathbf{L}_{-1} \mathbf{G}_4^2, \quad C_{p0} := 2\mathbf{R}_2 \mathbf{G}_4^2. \end{aligned}$$

Thus

$$C_b = \mathbf{S}(6)^{3/2} = \frac{2}{3^{3/4}\pi}, \quad C_{p0} = 2\mathbf{S}(6)^{3/2} = \frac{4}{3^{3/4}\pi}.$$

For every $0 < T < T_$,*

$$\|\mathcal{B}(u)\|_{L^{4/3}(0,T;H^{-1})}^{4/3} \leq \frac{C_b^{4/3}}{2\nu} \|u_0\|_2^{8/3}, \quad (54)$$

$$\|\mathcal{B}(u)\|_{L^2(0,T;\dot{H}^{-3/2})}^2 \leq \frac{\mathbf{S}(3)^6}{2\nu} \|u_0\|_2^4 = \frac{1}{4\pi^2\nu} \|u_0\|_2^4, \quad (55)$$

$$\|V_1\|_{L^2(0,T;\dot{H}^{-3/2})}^2 \leq \sqrt{2}\nu^{3/2}T^{1/2} \|u_0\|_2^2 + \frac{1}{2\pi^2\nu} \|u_0\|_2^4, \quad (56)$$

$$\|V_1\|_{L^{4/3}(0,T;H^{-1})}^{4/3} \leq 2^{1/3} \left(2^{-2/3}\nu^{2/3}T^{1/3} \|u_0\|_2^{4/3} + \frac{C_b^{4/3}}{2\nu} \|u_0\|_2^{8/3} \right), \quad (57)$$

$$\|p\|_{L^1(0,T;L^2)} + \|\nabla p\|_{L^1(0,T;H^{-1})} \leq 2^{-3/4}C_{p0}\nu^{-3/4}T^{1/4} \|u_0\|_2^2. \quad (58)$$

In the homogeneous base cell one also has, for every $t < T_$,*

$$\|V_1(t)\|_{\dot{H}^{-1}}^2 = \nu^2 \|u(t)\|_{\dot{H}^1}^2 + \|\mathcal{B}(u)(t)\|_{\dot{H}^{-1}}^2. \quad (59)$$

Define the nonlinear base action

$$\mathfrak{A}_{\text{nl}}(T) := \int_0^T \|\mathcal{B}(u)(t)\|_{\dot{H}^{-1}}^2 dt. \quad (60)$$

Then the homogeneous temporal cell has the exact action form

$$\|V_1\|_{L^2(0,T;\dot{H}^{-1})}^2 = \frac{\nu}{2} \left(\|u_0\|_2^2 - \|u(T)\|_2^2 \right) + \mathfrak{A}_{\text{nl}}(T). \quad (61)$$

Thus the datum-paid energy precursor

$$\mathbf{E}_{0,0}(T) := \left(\|u\|_{L^2(0,T;H^1)} \quad \|V_1\|_{L^{4/3}(0,T;H^{-1})} \quad \|p\|_{L^1(0,T;L^2)} + \|\nabla p\|_{L^1(0,T;H^{-1})} \right) \quad (62)$$

obeys the explicit coordinatewise estimate

$$\begin{aligned} \mathcal{E}_0(T) &:= \left(T + \frac{1}{2\nu} \right)^{1/2} \|u_0\|_2, \\ \mathcal{E}_1(T) &:= 2^{1/4} \left(2^{-2/3} \nu^{2/3} T^{1/3} \|u_0\|_2^{4/3} + \frac{C_b^{4/3}}{2\nu} \|u_0\|_2^{8/3} \right)^{3/4}, \\ \mathcal{E}_p(T) &:= 2^{-3/4} C_{p0} \nu^{-3/4} T^{1/4} \|u_0\|_2^2, \end{aligned} \quad (63)$$

and

$$\mathbf{E}_{0,0}(T) \preceq (\mathcal{E}_0(T) \quad \mathcal{E}_1(T) \quad \mathcal{E}_p(T)), \quad (64)$$

where $\preceq$ denotes coordinatewise inequality. The precursor is distinct from the rectangle matrix $\mathbf{C}_{0,0}(T)$: its middle coordinate has time exponent $4/3$, whereas $\mathbf{C}_{0,0}(T)$ has exponent 2.

Proof. Interpolation between L^2 and L^6 , followed by Lemma 1.3, gives

$$\|f\|_4 \leq \|f\|_2^{1/4} \|f\|_6^{3/4} \leq \mathfrak{S}(6)^{3/4} \|f\|_2^{1/4} \|\nabla f\|_2^{3/4}.$$

For $\xi \neq 0$, the symbol of $\mathbb{P}\nabla : H^{-1} \leftarrow L^2$ has tensor-to-vector norm $|\xi|/\langle \xi \rangle \leq 1$, and its supremum is one. The symbol $F \mapsto R_i R_j F_{ij}$ is contraction with $(\xi/|\xi|) \otimes (\xi/|\xi|)$, whose Frobenius norm is one. This proves the stated exact multiplier values.

The vector-valued Gagliardo–Nirenberg inequality and (52) give

$$\|\mathcal{B}(u)\|_{H^{-1}} \leq \mathfrak{L}_{-1} \|u \otimes u\|_2 \leq C_b \|u_0\|_2^{1/2} \|\nabla u\|_2^{3/2}.$$

Raise this inequality to the power $4/3$, integrate, and use (51); this proves (54). Dual Sobolev, the order $-1/2$ tensor multiplier, and Fourier log-convexity give

$$\begin{aligned} \|\mathcal{B}(u)\|_{\dot{H}^{-3/2}} &\leq \|u \otimes u\|_{\dot{H}^{-1/2}} \leq \mathfrak{S}(3) \|u \otimes u\|_{3/2} \\ &= \mathfrak{S}(3) \|u\|_3^2 \leq \mathfrak{S}(3)^3 \|u\|_{\dot{H}^{1/2}}^2 \leq \mathfrak{S}(3)^3 \|u\|_2 \|\nabla u\|_2. \end{aligned}$$

Squaring, integrating, and using $\mathfrak{S}(3)^6 = 1/(2\pi^2)$ and (51) prove (55). The projected momentum equation is

$$V_1 = \nu \Delta u - \mathcal{B}(u).$$

Moreover,

$$\int_0^T \|u\|_{\dot{H}^{1/2}}^2 dt \leq \|u_0\|_2 \int_0^T \|\nabla u\|_2 dt \leq \frac{T^{1/2}}{\sqrt{2\nu}} \|u_0\|_2^2.$$

Since $\|\Delta u\|_{\dot{H}^{-3/2}} = \|u\|_{\dot{H}^{1/2}}$, the square inequality for the projected equation and (55) prove (56). Since $\|\Delta u\|_{H^{-1}} \leq \|\nabla u\|_2$, Hölder's inequality and (51) give

$$\nu^{4/3} \int_0^T \|\Delta u\|_{H^{-1}}^{4/3} dt \leq 2^{-2/3} \nu^{2/3} T^{1/3} \|u_0\|_2^{4/3}.$$

Use $(a+b)^{4/3} \leq 2^{1/3}(a^{4/3} + b^{4/3})$ together with (54) to obtain (57).

Pressure normalization and the definition of $\mathbf{R}_2$ imply

$$\|p\|_2 + \|\nabla p\|_{H^{-1}} \leq 2\mathbf{R}_2 \|u \otimes u\|_2 \leq C_{p0} \|u_0\|_2^{1/2} \|\nabla u\|_2^{3/2}.$$

Here

$$\|\nabla p\|_{H^{-1}}^2 = \int_{\mathbb{R}^3} \frac{|\xi|^2}{1 + |\xi|^2} |\widehat{p}(\xi)|^2 d\xi \leq \|p\|_2^2.$$

Integration and

$$\int_0^T \|\nabla u\|_2^{3/2} dt \leq T^{1/4} \left(\int_0^T \|\nabla u\|_2^2 dt \right)^{3/4}$$

prove (58).

For the homogeneous identity, the Fourier multiplier of $\mathbb{P}\nabla \cdot$ gives

$$\|\mathcal{B}(u)\|_{\dot{H}^{-1}} \leq \|u \otimes u\|_2 < \infty.$$

Put $\mathcal{A}_S := -\Delta$ on solenoidal fields. Then

$$V_1 = -\nu \mathcal{A}_S u - \mathcal{B}(u), \quad \langle \mathcal{A}_S u, \mathcal{B}(u) \rangle_{\dot{H}^{-1}} = \langle u, \mathcal{B}(u) \rangle_{L^2} = 0.$$

Indeed, self-adjointness of $\mathbb{P}$, $\mathbb{P}u = u$, incompressibility, and integration by parts give

$$\langle u, \mathcal{B}(u) \rangle_{L^2} = -\frac{1}{2} \int_{\mathbb{R}^3} u \cdot \nabla |u|^2 dx = 0.$$

Expanding the square proves (59). Integrate that identity and use (51) to obtain (61). $\square$

Proposition 3.3 (Exact critical source action). *For $0 < T < T_*$, define*

$$\mathfrak{S}_{1/2}(T) := \int_0^T \|\mathcal{B}(u)(t)\|_{\dot{H}^{-1/2}}^2 dt \quad (65)$$

and

$$\mathcal{K}_{1/2}(T)^2 := \|u_0\|_{\dot{H}^{1/2}}^2 + \nu^{-1} \mathfrak{S}_{1/2}(T). \quad (66)$$

Then

$$\boxed{\nu \frac{d}{dt} \|u(t)\|_{\dot{H}^{1/2}}^2 + \nu^2 \|u(t)\|_{\dot{H}^{3/2}}^2 + \|V_1(t)\|_{\dot{H}^{-1/2}}^2 = \|\mathcal{B}(u)(t)\|_{\dot{H}^{-1/2}}^2} \quad (67)$$

for every $t < T_*$. Consequently,

$$\boxed{\sup_{0 \leq t \leq T} \left\{ \|u(t)\|_{\dot{H}^{1/2}}^2 + \nu \|u\|_{L^2(0,t;\dot{H}^{3/2})}^2 + \nu^{-1} \|V_1\|_{L^2(0,t;\dot{H}^{-1/2})}^2 \right\} \leq \mathcal{K}_{1/2}(T)^2.} \quad (68)$$

The inhomogeneous coordinates satisfy

$$\sup_{0 \leq t \leq T} \|u(t)\|_{H^{1/2}}^2 \leq \|u_0\|_2^2 + \mathcal{K}_{1/2}(T)^2, \quad (69)$$

$$\|u\|_{L^2(0,T;H^{3/2})}^2 \leq 2^{1/2} \left(T \|u_0\|_2^2 + \nu^{-1} \mathcal{K}_{1/2}(T)^2 \right), \quad (70)$$

$$\|V_1\|_{L^2(0,T;H^{-1})}^2 \leq \nu \mathcal{K}_{1/2}(T)^2. \quad (71)$$

Proof. Put

$$g := \Lambda^{-1/2} \mathcal{B}(u), \quad w := \Lambda^{-1/2} V_1 = -\nu \Lambda^{3/2} u - g.$$

The projected momentum equation gives

$$\frac{d}{dt} \|u\|_{\dot{H}^{1/2}}^2 = -2\nu \|u\|_{\dot{H}^{3/2}}^2 - 2 \operatorname{Re} \langle g, \Lambda^{3/2} u \rangle_2.$$

Multiplication by ν , followed by expansion of $\|w\|_2^2$, proves (67). Integrating it on $(0, t)$, using $\mathfrak{S}_{1/2}(t) \leq \mathfrak{S}_{1/2}(T)$, and taking the supremum proves (68).

The kinetic-energy identity pays the low-frequency piece of the inhomogeneous $H^{1/2}$ norm. Moreover,

$$(1 + |\xi|^2)^{3/2} \leq 2^{1/2} (1 + |\xi|^3), \quad (1 + |\xi|^2)^{-1} \leq |\xi|^{-1} \quad (\xi \neq 0).$$

These multiplier inequalities, (52), and (68) prove (69)–(71). $\square$

Definition 3.4 (Finite mixed rectangle). Let $0 < T < T_*$ and $N, M \in \mathbb{N}_0$. Define

$$\mathfrak{R}_{N,M}(u; T) := \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2, \quad (72)$$

$$\mathfrak{Q}_{N,M}(p; T) := \sum_{n=0}^N \left(\|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})} \right). \quad (73)$$

Set

$$\mathbf{R}_{N,M}[u_0; T] := \begin{pmatrix} V_0|_{(0,T)} & V_1|_{(0,T)} & Q_0|_{(0,T)} \\ V_1|_{(0,T)} & V_2|_{(0,T)} & Q_1|_{(0,T)} \\ \vdots & \vdots & \vdots \\ V_N|_{(0,T)} & V_{N+1}|_{(0,T)} & Q_N|_{(0,T)} \end{pmatrix}. \quad (74)$$

The three columns are read respectively in $L^2(0, T; H^{M+1})$, $L^2(0, T; H^{M-1})$, and $L^1(0, T; H^M)$, with the pressure-gradient coordinate retained through (73). The pressure-complete record $\mathcal{R}_{N,M}[u_0; T]$ is the object matrix (74) together with the velocity and pressure row identities in Proposition 2.4.

Sobolev rectangle matrix and adjacent stencil. For $m \in \mathbb{Z}$, set

$$G_{n,m}(T) := \|V_n\|_{L^2(0,T;H^m)},$$

and, for $M \in \mathbb{N}_0$, set

$$P_{n,M}(T) := \|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})}.$$

Applying the three column norms to (74) gives the scalar norm matrix

$$\mathbf{C}_{N,M}(T) := \begin{pmatrix} G_{0,M+1}(T) & G_{1,M-1}(T) & P_{0,M}(T) \\ G_{1,M+1}(T) & G_{2,M-1}(T) & P_{1,M}(T) \\ \vdots & \vdots & \vdots \\ G_{N,M+1}(T) & G_{N+1,M-1}(T) & P_{N,M}(T) \end{pmatrix}. \quad (75)$$

If $\mathbf{G}_{N,M}(T)$ denotes its first two columns, then

$$\|\mathbf{G}_{N,M}(T)\|_{\mathbb{F}}^2 = \mathfrak{R}_{N,M}(u; T), \quad \sum_{n=0}^N P_{n,M}(T) = \mathfrak{Q}_{N,M}(p; T). \quad (76)$$

For each $0 \leq n \leq N$, the first two entries in row n form the adjacent Hilbert-scale stencil

$$\left. \begin{array}{l} (n, M+1) : \quad V_n \in L^2(0, T; H^{M+1}), \\ (n+1, M-1) : \quad V_{n+1} = \partial_t V_n \in L^2(0, T; H^{M-1}) \end{array} \right\} \longrightarrow (n, M) : \quad V_n \in C([0, T]; H^M). \quad (77)$$

For every $m \in \mathbb{N}_0$, this Sobolev matrix measures the mixed-index array of Proposition 2.5 through the exact identity

$$G_{n,m}(T)^2 = \sum_{\substack{\alpha \in \mathbb{N}_0^3 \\ |\alpha| \leq m}} \frac{m!}{(m-|\alpha|)! \alpha!} \|J_{n,\alpha}\|_{L^2((0,T) \times \mathbb{R}^3)}^2. \quad (78)$$

The pressure entry in row n is rooted at $Q_n = \Pi_{n,0}$; its spatial rows $\partial_x^\alpha Q_n = \Pi_{n,\alpha}$ are fixed by (31).

Proof. The two identities in (76) are the definitions (72) and (73), arranged by temporal row. Lemma 1.7 gives (77). Finally, Plancherel and

$$(1 + |\xi|^2)^m = \sum_{\substack{\alpha \in \mathbb{N}_0^3 \\ |\alpha| \leq m}} \frac{m!}{(m-|\alpha|)! \alpha!} \xi^{2\alpha}$$

give (78). The pressure claims are the definitions of $Q_n, \Pi_{n,\alpha}$ and (31). These identities organize the finite coordinates; their whole-terminal bounds are supplied by Theorem 3.6 and Proposition 3.7. $\square$

Proposition 3.5 (Critical-action majorant for every finite rectangle). *Fix $0 < T < T_*$ and suppose that $\mathfrak{S}_{1/2}(T) < \infty$. Write*

$$\mathcal{N}(f, g) := \mathbb{P} \nabla \cdot (f \otimes g), \quad \mathcal{N}(v) := \mathcal{N}(v, v).$$

For integers $r \geq 1$, define

$$\mathsf{K}_r := \sum_{k=1}^r \binom{r}{k} 3^k \sum_{\ell=1}^k \binom{k}{\ell} \mathcal{S}(p_{k,\ell}) \mathcal{S}(q_{k,\ell}), \quad (79)$$

where $(p_{1,1}, q_{1,1}) = (3, 6)$, and, when $k \geq 2$,

$$\theta_{k,\ell} := \frac{\ell-1}{k-1}, \quad \frac{1}{p_{k,\ell}} := \frac{1}{3} - \frac{\theta_{k,\ell}}{6}, \quad \frac{1}{q_{k,\ell}} := \frac{1}{6} + \frac{\theta_{k,\ell}}{6}.$$

Then the solenoidal commutator estimate is

$$|\langle \mathcal{N}(v), v \rangle_{H^r}| \leq \mathsf{K}_r \|v\|_{\dot{H}^{3/2}} \|v\|_{H^r} \|\nabla v\|_{H^r}. \quad (80)$$

For $m \geq 0$, define the explicit order constant

$$\tilde{\mathbf{E}}_m := \max_{0 \leq j \leq m} C_{j+2,j}^{\text{emb}}, \quad \mathbf{M}_m := 13^{m/2} \tilde{\mathbf{E}}_m. \quad (81)$$

Then

$$\|\mathcal{N}(f, g)\|_{H^{m-1}} \leq \mathbf{M}_m \|f\|_{H^{m+2}} \|g\|_{H^m}, \quad (82)$$

$$\|\mathcal{N}(f, g)\|_{H^{m-1}} \leq \mathbf{M}_m \|g\|_{H^{m+2}} \|f\|_{H^m}. \quad (83)$$

All these constants are finite, explicit, and depend only on the displayed order and the dimension; $\mathbf{K}_r$ and $\mathbf{M}_m$ are given by the displayed Fourier–Sobolev formulas.

Set

$$\Theta_r(T) := \frac{\mathbf{K}_r^2}{\nu^2} \mathcal{K}_{1/2}(T)^2 \quad (r \geq 1). \quad (84)$$

Define

$$\mathcal{X}_{0,0} := \|u_0\|_2, \quad \mathcal{X}_{0,r} := \|u_0\|_{H^r} \exp\left(\frac{\Theta_r(T)}{2}\right) \quad (r \geq 1), \quad (85)$$

and

$$\begin{aligned} \mathcal{Y}_{0,0} &:= T^{1/2} \|u_0\|_2, \\ \mathcal{Y}_{0,1} &:= \left(T + \frac{1}{2\nu}\right)^{1/2} \|u_0\|_2, \\ \mathcal{Y}_{0,r} &:= (T + \nu^{-1})^{1/2} \|u_0\|_{H^{r-1}} \exp\left(\frac{\Theta_{r-1}(T)}{2}\right), \quad r \geq 2. \end{aligned} \quad (86)$$

The initial jet is majorized recursively by

$$\mathcal{A}_{0,m}^{\text{jet}} := \|u_0\|_{H^m}, \quad (87)$$

$$\mathcal{A}_{n+1,m}^{\text{jet}} := \nu \mathcal{A}_{n,m+2}^{\text{jet}} + \mathbf{M}_{m+1} \sum_{a=0}^n \binom{n}{a} \mathcal{A}_{a,m+3}^{\text{jet}} \mathcal{A}_{n-a,m+1}^{\text{jet}}. \quad (88)$$

For $n \geq 1$ and $m \geq 0$, define

$$\mathbf{F}_{1,m} := 0,$$

$$\mathbf{F}_{n,m} := \mathbf{M}_m \sum_{a=1}^{n-1} \binom{n}{a} \min\{\mathcal{X}_{a,m+2} \mathcal{Y}_{n-a,m}, \mathcal{X}_{n-a,m+2} \mathcal{Y}_{a,m}\}, \quad n \geq 2, \quad (89)$$

$$G_m(T) := 2\nu T + \frac{8\mathbf{M}_m^2}{\nu} \mathcal{Y}_{0,m+2}^2, \quad (90)$$

$$\mathcal{X}_{n,m} := \exp\left(\frac{G_m(T)}{2}\right) \left((\mathcal{A}_{n,m}^{\text{jet}})^2 + \frac{2}{\nu} \mathbf{F}_{n,m}^2\right)^{1/2}, \quad (91)$$

$$\mathcal{Y}_{n,0} := T^{1/2} \mathcal{X}_{n,0}, \quad (92)$$

$$\mathcal{Y}_{n,r} := \min\left\{T^{1/2} \mathcal{X}_{n,r}, \nu^{-1/2} \mathcal{X}_{n,r-1}\right\}, \quad r \geq 1. \quad (93)$$

The recursion is triangular in n , since $\mathbf{F}_{n,m}$ contains only rows of lower temporal order.

For the negative Sobolev edge, put

$$\mathbf{B}_{-1} := \mathbf{L}_{-1} \mathbf{G}_4^2, \quad \mathcal{Y}_{0,-1} := \mathcal{Y}_{0,0},$$

and, for $n \geq 1$, set

$$\mathcal{D}_{n,-1} := \nu \mathcal{Y}_{n-1,1} + \mathbf{B}_{-1} \sum_{a=0}^{n-1} \binom{n-1}{a} \mathcal{X}_{a,1} \mathcal{Y}_{n-1-a,1}. \quad (94)$$

Define

$$\mathcal{Y}_{1,-1} := \min \left\{ \mathcal{Y}_{1,0}, \mathcal{D}_{1,-1}, \nu^{1/2} \mathcal{K}_{1/2}(T) \right\}, \quad (95)$$

$$\mathcal{Y}_{n,-1} := \min \{ \mathcal{Y}_{n,0}, \mathcal{D}_{n,-1} \}, \quad n \geq 2. \quad (96)$$

Then, for every $n, m \in \mathbb{N}_0$,

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} \leq \mathcal{X}_{n,m}, \quad (97)$$

$$\|V_n\|_{L^2(0,T;H^m)} \leq \mathcal{Y}_{n,m}, \quad (98)$$

$$\|V_n\|_{L^2(0,T;H^{-1})} \leq \mathcal{Y}_{n,-1}. \quad (99)$$

For pressure, set

$$\lambda_0 := 1, \quad \lambda_1 := 2, \quad \lambda_m := m \quad (m \geq 2),$$

and set

$$\mathbf{P}_m := C_m^{\text{alg}}. \quad (100)$$

Then

$$\|R_i R_j(f_i g_j)\|_{H^m} \leq \mathbf{P}_m \|f\|_{H^{\lambda_m}} \|g\|_{H^{\lambda_m}}. \quad (101)$$

Define

$$\mathbf{Q}_{n,m}^{[\infty]} := \mathbf{P}_m \sum_{a=0}^n \binom{n}{a} \mathcal{X}_{a,\lambda_m} \mathcal{X}_{n-a,\lambda_m}, \quad (102)$$

$$\mathbf{Q}_{n,m}^{[2]} := \mathbf{P}_m \sum_{a=0}^n \binom{n}{a} \min \{ \mathcal{X}_{a,\lambda_m} \mathcal{Y}_{n-a,\lambda_m}, \mathcal{X}_{n-a,\lambda_m} \mathcal{Y}_{a,\lambda_m} \}. \quad (103)$$

Then

$$\|Q_n\|_{L^\infty(0,T;H^m)} + \|\nabla Q_n\|_{L^\infty(0,T;H^{m-1})} \leq 2\mathbf{Q}_{n,m}^{[\infty]}, \quad (104)$$

$$\|Q_n\|_{L^2(0,T;H^m)} + \|\nabla Q_n\|_{L^2(0,T;H^{m-1})} \leq 2\mathbf{Q}_{n,m}^{[2]}. \quad (105)$$

For $N, M \in \mathbb{N}_0$, the complete rectangle norm matrix satisfies

$$\mathbf{C}_{N,M}(T) \preceq \begin{pmatrix} \mathcal{Y}_{0,M+1} & \mathcal{Y}_{1,M-1} & 2T^{1/2} \mathbf{Q}_{0,M}^{[2]} \\ \mathcal{Y}_{1,M+1} & \mathcal{Y}_{2,M-1} & 2T^{1/2} \mathbf{Q}_{1,M}^{[2]} \\ \vdots & \vdots & \vdots \\ \mathcal{Y}_{N,M+1} & \mathcal{Y}_{N+1,M-1} & 2T^{1/2} \mathbf{Q}_{N,M}^{[2]} \end{pmatrix}. \quad (106)$$

In particular,

$$\mathfrak{R}_{N,M}(u; T) \leq \sum_{n=0}^N (\mathcal{Y}_{n,M+1}^2 + \mathcal{Y}_{n+1,M-1}^2), \quad (107)$$

$$\mathfrak{Q}_{N,M}(p; T) \leq 2T^{1/2} \sum_{n=0}^N \mathfrak{Q}_{n,M}^{[2]}. \quad (108)$$

Every entry on the right side of (106) is a finite recursion in

$$u_0, \quad \nu, \quad T, \quad \mathfrak{S}_{1/2}(T),$$

and the displayed dimension/order constants. This proposition is a conditional quantitative readout of the source action. It neither supplies a datum-only whole-terminal action bound nor replaces Theorem 3.6, whose datum-generated terminal statement remains the rectangle producer.

Proof. For (80), expansion of $(1 + |\xi|^2)^r$, self-adjointness of $\mathbb{P}$, and incompressibility give, for every integer $r \geq 1$,

$$\langle A^r \mathcal{N}(v), A^r v \rangle_{L^2} = \sum_{|\alpha| \leq r} \frac{r!}{(r - |\alpha|)! \alpha!} \langle \partial^\alpha ((v \cdot \nabla) v), \partial^\alpha v \rangle_{L^2}.$$

The undifferentiated transport term in each Leibniz expansion has zero real pairing. A nonzero remaining term has $k = |\alpha|$, $\ell = |\beta|$, and is bounded by

$$\left\| \partial^\beta v \right\|_{p_{k,\ell}} \left\| \nabla \partial^{\alpha-\beta} v \right\|_{q_{k,\ell}} \left\| \partial^\alpha v \right\|_2.$$

For $k \geq 2$, the exponents above satisfy

$$\begin{aligned} \ell + s(p_{k,\ell}) &= (1 - \theta_{k,\ell}) \frac{3}{2} + \theta_{k,\ell} (k + 1), \\ k - \ell + 1 + s(q_{k,\ell}) &= \theta_{k,\ell} \frac{3}{2} + (1 - \theta_{k,\ell}) (k + 1). \end{aligned}$$

Lemma 1.3 and Fourier log-convexity thus give

$$\begin{aligned} &\left\| \partial^\beta v \right\|_{p_{k,\ell}} \left\| \nabla \partial^{\alpha-\beta} v \right\|_{q_{k,\ell}} \\ &\leq \mathfrak{S}(p_{k,\ell}) \mathfrak{S}(q_{k,\ell}) \|v\|_{\dot{H}^{3/2}} \|v\|_{\dot{H}^{k+1}}. \end{aligned}$$

For $k = 1$, the same line is the direct L^3 - L^6 estimate. Since $k \leq r$,

$$\|v\|_{\dot{H}^{k+1}} \leq \|\nabla v\|_{H^r}, \quad \|\partial^\alpha v\|_2 \leq \|v\|_{H^r}.$$

Finally,

$$\sum_{|\alpha|=k} \frac{r!}{(r-k)! \alpha!} = \binom{r}{k} 3^k, \quad \sum_{\substack{\beta \leq \alpha \\ |\beta|=\ell}} \binom{\alpha}{\beta} = \binom{k}{\ell}.$$

Summing proves the commutator estimate with the explicit constant (79). To prove (82), use

$$\|\mathbb{P} \nabla \cdot F\|_{H^{m-1}} \leq \|F\|_{H^m}.$$

For every $|\alpha| \leq m$, the Leibniz rule and the embedding $H^{j+2} \hookrightarrow W^{j,\infty}$ give

$$\|\partial^\alpha (f_i g_j)\|_2 \leq \tilde{\mathfrak{E}}_m 2^{|\alpha|} \|f_i\|_{H^{m+2}} \|g_j\|_{H^m}.$$

The exact integer Sobolev identity from Lemma 1.4, followed by

$$\sum_{|\alpha| \leq m} \frac{m!}{(m - |\alpha|)! \alpha!} 4^{|\alpha|} = 13^m,$$

proves (82) with the stated $\mathbf{M}_m$. Interchanging f and g proves (83). The H^r energy row and Young's inequality give

$$\frac{d}{dt} \|u\|_{H^r}^2 + \nu \|\nabla u\|_{H^r}^2 \leq \frac{K_r^2}{\nu} \|u\|_{\dot{H}^{3/2}}^2 \|u\|_{H^r}^2.$$

Equations (68) and Gronwall give

$$\sup_{0 \leq t \leq T} \|u(t)\|_{H^r}^2 \leq \|u_0\|_{H^r}^2 e^{\Theta_r(T)}, \quad \int_0^T \|\nabla u\|_{H^r}^2 dt \leq \nu^{-1} \|u_0\|_{H^r}^2 e^{\Theta_r(T)}.$$

Together with $\|f\|_{H^{r+1}}^2 = \|f\|_{H^r}^2 + \|\nabla f\|_{H^r}^2$, the kinetic estimates prove (85)–(86).

Evaluation of the projected temporal recurrence at $t = 0$, followed by (82), proves

$$\|V_n(0)\|_{H^m} \leq \mathcal{A}_{n,m}^{\text{jet}}$$

by induction. For $n \geq 1$, write

$$\partial_t V_n - \nu \Delta V_n + \mathcal{N}(u, V_n) + \mathcal{N}(V_n, u) + F_n = 0,$$

where

$$F_n := \sum_{a=1}^{n-1} \binom{n}{a} \mathcal{N}(V_a, V_{n-a}).$$

The induction hypothesis and (82)–(83) give

$$\|F_n\|_{L^2(0,T;H^{m-1})} \leq F_{n,m}.$$

Pairing the row in H^m , using the two mixed product estimates, and absorbing the H^{m+1} factors gives

$$\begin{aligned} \frac{d}{dt} \|V_n\|_{H^m}^2 + \nu \|V_n\|_{H^{m+1}}^2 &\leq \left(2\nu + \frac{8M_m^2}{\nu} \|u\|_{H^{m+2}}^2 \right) \|V_n\|_{H^m}^2 \\ &\quad + \frac{2}{\nu} \|F_n\|_{H^{m-1}}^2. \end{aligned} \tag{109}$$

The integrating-factor form of (109) proves (91) and

$$\nu \|V_n\|_{L^2(0,T;H^{m+1})}^2 \leq \mathcal{X}_{n,m}^2.$$

This proves (97)–(98) by temporal induction.

The projected recurrence at order $n - 1$, the bilinear estimate

$$\|\mathcal{N}(f, g)\|_{H^{-1}} \leq \mathbf{B}_{-1} \|f\|_{H^1} \|g\|_{H^1},$$

and Hölder in time prove $\|V_n\|_{L^2 H^{-1}} \leq \mathcal{D}_{n,-1}$. Equation (71) supplies the third bound in (95). This proves (99).

For the pressure product, the multiplier $F \mapsto R_i R_j F_{ij}$ has operator norm at most one from the componentwise tensor H^m norm to H^m . If $m = 0$, use the first estimate of Lemma 1.4 at order zero; if $m = 1$, use it at order one; and if $m \geq 2$, use the algebra estimate. These cases give (101) with $\mathbf{P}_m = C_m^{\text{alg}}$. The pressure recurrence and that estimate give the two bounds for Q_n . The multiplier inequality $\|\nabla q\|_{H^{m-1}} \leq \|q\|_{H^m}$ gives the gradient bounds. Finally, $L^2(0, T) \hookrightarrow L^1(0, T)$ with norm $T^{1/2}$, the definitions of $\mathbf{C}_{N,M}(T)$, $\mathfrak{R}_{N,M}$, and $\mathfrak{Q}_{N,M}$, and the special negative-edge recursion for $M = 0$ prove (106)–(108). $\square$

Theorem 3.6 (Datum-generated whole-terminal rectangles). *Let $\nu > 0$, $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, and let (T_*, u, p) be the maximal pressure-complete classical history generated by (u_0, ν) . If $T_* < \infty$, then, for every $N, M \in \mathbb{N}_0$,*

$$\mathfrak{R}_{N,M}^*[u_0, \nu, T_*] := \sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty. \quad (110)$$

Equivalently, for every $n, m \in \mathbb{N}_0$,

$$V_n \in L^2(0, T_*; H^{m+1}), \quad V_{n+1} \in L^2(0, T_*; H^{m-1}). \quad (111)$$

Every coordinate in (111) is the corresponding distributional derivative of the single maximal history.

Proposition 3.7 (Pressure completion). *Under the hypotheses of Theorem 3.6,*

$$\mathfrak{Q}_{N,M}^*[u_0, \nu, T_*] := \sup_{0 < T < T_*} \mathfrak{Q}_{N,M}(p; T) \leq C_M^p (2^{N+1} - 1) \mathfrak{R}_{N,M}^*[u_0, \nu, T_*]. \quad (112)$$

For every $0 \leq n \leq N$, the identities

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}, \quad (113)$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n \quad (114)$$

hold respectively in $L^1(0, T_; H^M)$ and $L^1(0, T_*; H^{M-1})$, including when $M = 0$.*

Proof. For every $0 < T < T_*$, Theorem 2.9 gives

$$\|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})} \leq C_M^p 2^n \mathfrak{R}_{N,M}^*[u_0, \nu, T_*].$$

Sum over $0 \leq n \leq N$, then let $T \uparrow T_*$. Monotone convergence in the nonnegative norm integrals proves (112). Since $\nabla \cdot V_a = 0$,

$$(V_a \cdot \nabla) V_{n-a} = \nabla \cdot (V_a \otimes V_{n-a}).$$

Lemma 1.4 and Cauchy–Schwarz in time give

$$\|V_a \otimes V_{n-a}\|_{L^1(0,T_*;H^M)} \leq C_M^{\text{alg}} \|V_a\|_{L^2(0,T_*;H^{M+1})} \|V_{n-a}\|_{L^2(0,T_*;H^{M+1})}.$$

Thus every transport term belongs to $L^1(0, T_*; H^{M-1})$, also for $M = 0$. Moreover,

$$\|\Delta V_n\|_{L^1 H^{M-1}} \leq T_*^{1/2} \|V_n\|_{L^2 H^{M+1}}.$$

Theorem 3.6 places V_{n+1} in $L^2(0, T_*; H^{M-1})$, hence in $L^1(0, T_*; H^{M-1})$. The strict-interval identities are identities of distributions; their whole-terminal representatives belong to the stated spaces. This proves the two identities on $(0, T_*)$. $\square$

Definition 3.8 (Finite-window spaces). For $N, M \in \mathbb{N}_0$, set

$$\mathcal{X}_{N,M} := \prod_{n=0}^N L^2(0, T_*; H^{M+1}) \times \prod_{n=0}^N L^2(0, T_*; H^{M-1}) \times \prod_{n=0}^N L^1(0, T_*; H^M).$$

The whole-terminal rectangle is the element

$$\mathcal{R}_{N,M}[u_0; T_*] := ((V_n)_{n=0}^N, (V_{n+1})_{n=0}^N; (Q_n)_{n=0}^N) \in \mathcal{X}_{N,M}. \quad (115)$$

It is represented by the object matrix $\mathbf{R}_{N,M}[u_0; T_*]$ from (74), with every entry understood on $(0, T_*)$. Define its datum fibre by

$$\mathcal{D}_{N,M}[u_0; T_*] := \{\mathcal{R}_{N,M}[u_0; T_*]\} \subset \mathcal{X}_{N,M}. \quad (116)$$

If $N' \leq N$ and $M' \leq M$, let

$$\rho_{N',M'}^{N,M} : \mathcal{X}_{N,M} \longrightarrow \mathcal{X}_{N',M'}$$

delete the rows with index larger than N' in $\mathbf{R}_{N,M}[u_0; T_*]$ and apply, column by column,

$$H^{M+1} \hookrightarrow H^{M'+1}, \quad H^{M-1} \hookrightarrow H^{M'-1}, \quad H^M \hookrightarrow H^{M'}$$

to the retained coordinates.

Proposition 3.9 (Restriction compatibility). *For $N' \leq N$ and $M' \leq M$,*

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0; T_*] = \mathcal{R}_{N',M'}[u_0; T_*]. \quad (117)$$

If $(N'', M'') \leq (N', M') \leq (N, M)$, then

$$\rho_{N'',M''}^{N',M'} \rho_{N',M'}^{N,M} = \rho_{N'',M''}^{N,M}.$$

Thus the restrictions of the maps $\rho_{N',M'}^{N,M}$ to the datum fibres form an inverse system.

Proof. All retained entries are the same distributions $\partial_t^n u$, $\partial_t^{n+1} u$, and $\partial_t^n p$. The Sobolev embeddings preserve those distributions. Deletion of rows and composition of canonical embeddings are associative. $\square$

Definition 3.10 (Compatible intersection).

$$\mathcal{I}[u_0] := \varprojlim_{N,M} \mathcal{D}_{N,M}[u_0; T_*] \quad (118)$$

is the subset of $\prod_{N,M \in \mathbb{N}_0} \mathcal{D}_{N,M}[u_0; T_*]$ given by

$$\mathcal{I}[u_0] = \left\{ (X_{N,M})_{N,M} : \rho_{N',M'}^{N,M} X_{N,M} = X_{N',M'} \text{ whenever } N' \leq N, M' \leq M \right\}. \quad (119)$$

Proposition 3.11 (Mixed Sobolev realization). *The velocity coordinate of $\mathcal{I}[u_0]$ satisfies*

$$u \in \bigcap_{r,m \in \mathbb{N}_0} H^r(0, T_*; H^m(\mathbb{R}^3)). \quad (120)$$

For every $r, m \in \mathbb{N}_0$,

$$\sum_{n=0}^r \|\partial_t^n u\|_{L^2(0, T_*; H^m)}^2 \leq \mathfrak{R}_{r,m}^*[u_0, \nu, T_*]. \quad (121)$$

Proof. Fix r, m . The rectangle $(N, M) = (r, m)$ contains

$$\partial_t^n u \in L^2(0, T_*; H^{m+1}) \hookrightarrow L^2(0, T_*; H^m) \quad (0 \leq n \leq r).$$

These entries are the distributional time derivatives of u , so $u \in H^r(0, T_*; H^m)$. The norm bound follows from (110) and Sobolev nesting. $\square$

Definition 3.12 (Critical-height rows). For $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t),$$

$$\mathcal{P}_s(t) := -\operatorname{Re} \langle \Lambda^s u(t), \Lambda^s((u \cdot \nabla)u + \nabla p)(t) \rangle_{L^2}.$$

Proposition 3.13 (Completed height recurrence). For every $0 < T < T_*$, $s \geq 0$, and $t \in (0, T)$,

$$E'_s = -2\nu E_{s+1} + \mathcal{P}_s, \quad (122)$$

and, for every $n \in \mathbb{N}$ and $t \in (0, T)$,

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}. \quad (123)$$

Consequently, for every $n \in \mathbb{N}$ and $t \in (0, T)$,

$$\frac{H^{(n)} - \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}}{(-2\nu)^n} = E_{n+1/2}. \quad (124)$$

Proof. Pair the momentum equation with $\Lambda^{2s}u$. Self-adjointness of Λ and $\langle \Lambda^s u, \Lambda^s \Delta u \rangle_{L^2} = -\|\Lambda^{s+1}u\|_2^2$ give (122). We first prove, for every $n \in \mathbb{N}$,

$$\partial_t^n E_s = (-2\nu)^n E_{s+n} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{s+j}. \quad (125)$$

For $n = 1$, this is (122). Suppose it holds at order n . Differentiation gives

$$\partial_t^{n+1} E_s = (-2\nu)^n \partial_t E_{s+n} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-j} \mathcal{P}_{s+j}.$$

Substitution of $\partial_t E_{s+n} = -2\nu E_{s+n+1} + \mathcal{P}_{s+n}$ yields

$$\partial_t^{n+1} E_s = (-2\nu)^{n+1} E_{s+n+1} + \sum_{j=0}^n (-2\nu)^j \partial_t^{n-j} \mathcal{P}_{s+j},$$

which is (125) at order $n+1$. Taking $s = 1/2$ proves (123), and division by $(-2\nu)^n$ proves (124). $\square$

4 Compatible intersection, endpoint jet, and restart

Theorem 4.1 (Compatible inverse intersection). Let $\nu > 0$, $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, and let (T_*, u, p) be the maximal pressure-complete classical history generated by (u_0, ν) . If $T_* < \infty$, then the family

$$(\mathcal{R}_{N,M}[u_0; T_*])_{N,M \in \mathbb{N}_0} \quad (126)$$

is the unique datum-generated element of the compatible intersection $\mathcal{I}[u_0]$ from Definition 3.10. It has unique coordinates $(V_n, Q_n)_{n \in \mathbb{N}_0}$ satisfying

$$V_n \in \bigcap_{m \in \mathbb{N}_0} L^2(0, T_*; H_\sigma^m(\mathbb{R}^3)), \quad (127)$$

$$Q_n \in \bigcap_{m \in \mathbb{N}_0} L^1(0, T_*; H^m(\mathbb{R}^3)), \quad (128)$$

and every finite restriction of these coordinates is the corresponding rectangle $\mathcal{R}_{N,M}[u_0; T_*]$.

Proof. Theorem 3.6 places every velocity coordinate of $\mathcal{R}_{N,M}[u_0; T_*]$ in its whole-terminal adjacent spaces. Proposition 3.7 places every retained pressure coordinate in its whole-terminal pressure space. Proposition 3.9 gives

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0; T_*] = \mathcal{R}_{N',M'}[u_0; T_*] \quad (N' \leq N, M' \leq M),$$

so (126) belongs to $\mathcal{I}[u_0]$. Equivalently, the object matrices $\mathbf{R}_{N,M}[u_0; T_*]$ in (74) agree under row truncation and the three column embeddings. Their compatible finite windows are the matrix coordinates of this inverse intersection.

Fix n . Every rectangle with $N \geq n$ contains the same distribution $\partial_t^n u$, and every such rectangle contains the same distribution $\partial_t^n p$. Increasing M , using the canonical Sobolev embeddings, gives (127) and (128). Any datum-generated compatible family has these same finite restrictions by Proposition 3.9; equality in every finite restriction gives equality of every distributional coordinate. $\square$

Corollary 4.2 (Whole-terminal pressure-complete mixed jet). *Under the hypotheses of Theorem 4.1, for every $r, m, n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$u \in H^r(0, T_*; H_\sigma^m(\mathbb{R}^3)), \quad (129)$$

$$J_{n,\alpha} := \partial_x^\alpha V_n \in L^2(0, T_*; H^{m+1}(\mathbb{R}^3)), \quad (130)$$

$$J_{n+1,\alpha} \in L^2(0, T_*; H^{m-1}(\mathbb{R}^3)), \quad (131)$$

$$\Pi_{n,\alpha} := \partial_x^\alpha Q_n \in L^1(0, T_*; H^m(\mathbb{R}^3)). \quad (132)$$

The following finite bounds hold:

$$\sum_{j=0}^r \|V_j\|_{L^2(0, T_*; H^m)}^2 \leq \mathfrak{R}_{r,m}^*[u_0, \nu, T_*], \quad (133)$$

$$\|J_{n,\alpha}\|_{L^2(0, T_*; H^{m+1})}^2 + \|J_{n+1,\alpha}\|_{L^2(0, T_*; H^{m-1})}^2 \leq \mathfrak{R}_{n,m+|\alpha|}^*[u_0, \nu, T_*], \quad (134)$$

$$\|\Pi_{n,\alpha}\|_{L^1(0, T_*; H^m)} \leq C_{m+|\alpha|}^p (2^{n+1} - 1) \mathfrak{R}_{n,m+|\alpha|}^*[u_0, \nu, T_*]. \quad (135)$$

Every mixed row satisfies the recurrences in Proposition 2.4 on $(0, T_*)$.

Proof. Equation (129) and (133) are Proposition 3.11. The Fourier multiplier bound

$$\|\partial_x^\alpha f\|_{H^s} \leq \|f\|_{H^{s+|\alpha|}}$$

and the rectangle $(n, m + |\alpha|)$ give (130)–(134). Apply the same multiplier bound to the pressure completion in Proposition 3.7 to obtain (132) and (135). Proposition 2.4 supplies the stated identities. $\square$

Theorem 4.3 (Compatible whole-terminal Sobolev traces). *Assume the hypotheses of Theorem 4.1. For $N, m \in \mathbb{N}_0$, define*

$$\mathfrak{E}_{N,m} := (1 + T_*^{-1})^{1/2} (\mathfrak{R}_{N,m}^*[u_0, \nu, T_*])^{1/2}. \quad (136)$$

If $0 \leq n \leq N$ and $\alpha \in \mathbb{N}_0^3$, then

$$V_n \in C([0, T_*]; H_\sigma^m(\mathbb{R}^3)), \quad \sup_{0 \leq t \leq T_*} \|V_n(t)\|_{H^m} \leq \mathfrak{E}_{N,m}, \quad (137)$$

$$J_{n,\alpha} \in C([0, T_*]; H^m(\mathbb{R}^3)), \quad \sup_{0 \leq t \leq T_*} \|J_{n,\alpha}(t)\|_{H^m} \leq \mathfrak{E}_{N,m+|\alpha|}. \quad (138)$$

There are unique values

$$V_n^* := \lim_{t \uparrow T_*} V_n(t) \in H^\infty(\mathbb{R}^3), \quad (139)$$

$$J_{n,\alpha}^* := \lim_{t \uparrow T_*} J_{n,\alpha}(t) = \partial_x^\alpha V_n^* \in H^\infty(\mathbb{R}^3), \quad (140)$$

such that, for $k \geq m$, the image of the H^k trace under $H^k \hookrightarrow H^m$ is the H^m trace. Quantitatively,

$$\|V_n^*\|_{H^m} \leq \mathfrak{E}_{n,m}, \quad (141)$$

$$\|V_n(t) - V_n^*\|_{H^m} \leq (T_* - t) \mathfrak{E}_{n+1,m}, \quad 0 \leq t \leq T_*, \quad (142)$$

$$\|J_{n,\alpha}^*\|_{H^m} \leq \mathfrak{E}_{n,m+|\alpha|}, \quad (143)$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}^*\|_{H^m} \leq (T_* - t) \mathfrak{E}_{n+1,m+|\alpha|}. \quad (144)$$

In particular,

$$u_* := V_0^* \in H_\sigma^\infty(\mathbb{R}^3), \quad u \in \bigcap_{a,m \in \mathbb{N}_0} C^a([0, T_*]; H_\sigma^m(\mathbb{R}^3)), \quad \partial_t^a u(T_*) = V_a^*. \quad (145)$$

Proof. For $0 \leq n \leq N$, Theorem 3.6 gives

$$V_n \in L^2(0, T_*; H^{m+1}), \quad \partial_t V_n = V_{n+1} \in L^2(0, T_*; H^{m-1}).$$

Lemma 1.7 gives the first membership in (137) and

$$\begin{aligned} \sup_{0 \leq t \leq T_*} \|V_n(t)\|_{H^m}^2 &\leq T_*^{-1} \|V_n\|_{L^2 H^m}^2 + 2 \|V_n\|_{L^2 H^{m+1}} \|V_{n+1}\|_{L^2 H^{m-1}} \\ &\leq (1 + T_*^{-1}) \mathfrak{R}_{N,m}^*[u_0, \nu, T_*]. \end{aligned}$$

This proves the bound in (137). Apply the same lemma to $J_{n,\alpha}$, using (134), to prove (138).

For $k \geq m$, the H^k -continuous and H^m -continuous representatives equal the same preterminal distribution. After applying $H^k \hookrightarrow H^m$, uniqueness of continuous representatives makes them equal on $[0, T_*]$. Their endpoint values are compatible and define (139)–(140). Spatial differentiation is continuous between the corresponding finite Sobolev spaces, so $J_{n,\alpha}^* = \partial_x^\alpha V_n^*$.

Corollary 4.2 gives $V_n, V_{n+1} \in L^1(0, T_*; H^m)$ and $\partial_t V_n = V_{n+1}$ in distributions. The Banach-valued fundamental identity gives

$$V_n(t) - V_n(s) = \int_s^t V_{n+1}(\tau) d\tau \quad \text{in } H^m, \quad 0 \leq s \leq t \leq T_*. \quad (146)$$

The continuous representative of V_{n+1} is bounded by $\mathfrak{E}_{n+1,m}$. Set $t = T_*$ in (146) to prove (142); spatial differentiation gives (144). The remaining size estimates follow from (137)–(138).

The divergence map $H^{m+1} \rightarrow H^m$ is continuous, so $\nabla \cdot u_* = 0$. For each a, m , the curves $V_0, \dots, V_{a+1}$ are continuous in H^m , and (146) identifies the derivative of V_j with V_{j+1} . Induction in j proves (145). $\square$

Theorem 4.4 (Pressure-complete endpoint jet). *Assume the hypotheses of Theorem 4.1. For $n \in \mathbb{N}_0$, define*

$$Q_n^* := R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}^* V_{n-a,j}^*. \quad (147)$$

Then $Q_n^* \in H^\infty(\mathbb{R}^3)$, and, for every $s \in \mathbb{N}_0$,

$$\|Q_n^*\|_{H^s} \leq 9C_s^{\text{alg}} 2^n \mathfrak{E}_{n,s+1}^2, \quad (148)$$

$$\|Q_n(t) - Q_n^*\|_{H^s} \leq 9C_s^{\text{alg}} 2^{n+1} (T_* - t) \mathfrak{E}_{n+1,s+1}^2. \quad (149)$$

For $\alpha \in \mathbb{N}_0^3$,

$$\Pi_{n,\alpha}^* := \partial_x^\alpha Q_n^* = \lim_{t \uparrow T_*} \Pi_{n,\alpha}(t) \quad \text{in } H^m(\mathbb{R}^3) \text{ for every } m \in \mathbb{N}_0. \quad (150)$$

More precisely, for every $m \in \mathbb{N}_0$,

$$\|\Pi_{n,\alpha}^*\|_{H^m} \leq 9C_{m+|\alpha|}^{\text{alg}} 2^n \mathfrak{E}_{n,m+|\alpha|+1}^2, \quad (151)$$

$$\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}^*\|_{H^m} \leq 9C_{m+|\alpha|}^{\text{alg}} 2^{n+1} (T_* - t) \mathfrak{E}_{n+1,m+|\alpha|+1}^2. \quad (152)$$

The endpoint rows satisfy, for every $n, q \in \mathbb{N}_0$,

$$V_{n+1}^* = \nu \Delta V_n^* - \sum_{a=0}^n \binom{n}{a} (V_a^* \cdot \nabla) V_{n-a}^* - \nabla Q_n^* \quad \text{in } H^q(\mathbb{R}^3). \quad (153)$$

The family $(V_n^*, Q_n^*)_{n \in \mathbb{N}_0}$ is the unique pressure-normalized endpoint jet rooted at u_* . Moreover,

$$p \in \bigcap_{a,m \in \mathbb{N}_0} C^a([0, T_*]; H^m(\mathbb{R}^3)), \quad \partial_t^a p(T_*) = Q_a^*. \quad (154)$$

Proof. Lemmas 1.4 and 1.5, followed by (137), give

$$\begin{aligned} \|Q_n^*\|_{H^s} &\leq 9C_s^{\text{alg}} \sum_{a=0}^n \binom{n}{a} \|V_a^*\|_{H^{s+1}} \|V_{n-a}^*\|_{H^{s+1}} \\ &\leq 9C_s^{\text{alg}} 2^n \mathfrak{E}_{n,s+1}^2. \end{aligned}$$

For each product in $Q_n(t) - Q_n^*$, insert $V_a^* V_{n-a}(t)$, use (142), and bound every retained factor by $\mathfrak{E}_{n+1,s+1}$. The two product differences and $\sum_{a=0}^n \binom{n}{a} = 2^n$ prove (149). The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^m} \leq \|f\|_{H^{m+|\alpha|}}$$

applied to (148) and (149) proves (151) and (152). The latter tends to zero as $t \uparrow T_*$, which proves (150) in every H^m .

Fix n, q . The trace of V_n in H^{q+2} carries the diffusion term in Proposition 2.4 to H^q at T_* . The product estimate at order $q+1$, using velocity convergence in H^{q+2} , carries every transport term to H^q . Equation (149) at order $q+1$ carries the pressure gradient to H^q . The left side converges to V_{n+1}^* in H^q , which proves (153).

The value $V_0^* = u_*$ fixes Q_0^* , and the velocity recurrence fixes V_1^* . Inductively, $V_0^*, \dots, V_n^*$ fix Q_n^* , which fixes V_{n+1}^* . This proves uniqueness. Extend each Q_n to T_* by $Q_n(T_*) := Q_n^*$. On $(0, T_*)$, $Q_n = \partial_t^n p$, so $\partial_t Q_n = Q_{n+1}$ in distributions. Both rows are continuous in every H^m by (149). Hence, for $0 \leq s \leq t \leq T_*$,

$$Q_n(t) - Q_n(s) = \int_s^t Q_{n+1}(\tau) d\tau \quad \text{in } H^m(\mathbb{R}^3). \quad (155)$$

Indeed, the identity holds for $t < T_*$, and the endpoint bounds permit passage to $t = T_*$. Thus $Q_n \in C^1([0, T_*]; H^m)$ with derivative Q_{n+1} . Induction gives (154). $\square$

Lemma 4.5 (Late-slice overlap uniqueness). *Assume the hypotheses of Theorem 4.1. Define*

$$\delta_0 := \min\{1, c_\nu(1 + \mathfrak{E}_{0,3})^{-2}\} > 0, \quad (156)$$

where c_ν is the constant in Theorem 1.10. For every $t_0 \in (\max\{0, T_* - \delta_0/2\}, T_*)$, let (w^{t_0}, q^{t_0}) be the pressure-normalized local solution generated by $u(t_0)$, translated to $[t_0, t_0 + \delta_0]$. Let (w^*, q^*) be the pressure-normalized local solution generated by u_* , translated to $[T_*, T_* + \delta_0]$. Then

$$(w^{t_0}, q^{t_0}) = (u, p) \quad \text{on } [t_0, T_*], \quad (157)$$

$$(w^{t_0}, q^{t_0}) = (w^*, q^*) \quad \text{on } [T_*, t_0 + \delta_0]. \quad (158)$$

Proof. Equation (137) gives

$$\|u(t_0)\|_{H^3} \leq \mathfrak{E}_{0,3}, \quad \|u_*\|_{H^3} \leq \mathfrak{E}_{0,3}.$$

Theorem 1.10, translated in time, gives both local solutions on the stated intervals. Its strong uniqueness gives (157). Continuity in H^3 gives

$$w^{t_0}(T_*) = \lim_{t \uparrow T_*} u(t) = u_*.$$

The two local solutions starting at T_* have the same datum and the same pressure normalization, so strong uniqueness gives (158). The right overlap has positive length because $t_0 + \delta_0 > T_* + \delta_0/2$. $\square$

Theorem 4.6 (Same-history endpoint restart). *Assume the hypotheses of Theorem 4.1. There is a pressure-normalized classical solution $(\tilde{u}, \tilde{p})$ on $[0, T_* + \delta_0/2] \times \mathbb{R}^3$ such that*

$$(\tilde{u}, \tilde{p})|_{[0, T_*] \times \mathbb{R}^3} = (u, p), \quad (159)$$

$$\tilde{u}, \tilde{p} \in C^\infty([0, T_* + \delta_0/2] \times \mathbb{R}^3), \quad (160)$$

$$\partial_t^n \tilde{u}(T_*) = V_n^*, \quad \partial_t^n \tilde{p}(T_*) = Q_n^* \quad (n \in \mathbb{N}_0). \quad (161)$$

The extension is the unique local continuation of the same maximal history.

Proof. Let (w^*, q^*) be the endpoint solution from Lemma 4.5, and set

$$W_n^+ := \partial_t^n w^*(T_*), \quad P_n^+ := \partial_t^n q^*(T_*).$$

Theorem 1.10 and Proposition 2.4 give

$$P_n^+ = R_i R_j \sum_{a=0}^n \binom{n}{a} W_{a,i}^+ W_{n-a,j}^+,$$

$$W_{n+1}^+ = \nu \Delta W_n^+ - \sum_{a=0}^n \binom{n}{a} (W_a^+ \cdot \nabla) W_{n-a}^+ - \nabla P_n^+.$$

Since $W_0^+ = u_* = V_0^*$, induction with (147) and (153) gives

$$W_n^+ = V_n^*, \quad P_n^+ = Q_n^*, \quad n \in \mathbb{N}_0. \quad (162)$$

Define

$$(\tilde{u}, \tilde{p})(t) := \begin{cases} (u, p)(t), & 0 \leq t < T_*, \\ (w^*, q^*)(t), & T_* \leq t \leq T_* + \delta_0/2. \end{cases}$$

Equations (145), (154), and (162) match every one-sided time derivative in every finite Sobolev space. Sobolev embedding gives (160), and the pressure-complete equation holds at T_* by (153). Lemma 4.5 identifies the joined pair near T_* with the single local history (w^{t_0}, q^{t_0}) . This proves (159) and local uniqueness of the continuation. $\square$

Corollary 4.7 (Maximal-time contradiction). *The maximal pressure-complete classical history generated by (u_0, ν) cannot have $T_* < \infty$.*

Proof. If $T_* < \infty$, Theorem 3.6 supplies the settled whole-terminal rectangles. Theorems 4.1–4.6 produce a pressure-normalized classical extension to $[0, T_* + \delta_0/2]$, where $\delta_0 > 0$. This contradicts maximality in Definition 2.2. $\square$

Theorem 4.8 (Global smoothness on $\mathbb{R}^3$). *For every $\nu > 0$ and every $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, there exists a pressure-normalized pair*

$$(u, p) \in C^\infty([0, \infty) \times \mathbb{R}^3; \mathbb{R}^3 \times \mathbb{R}) \quad (163)$$

satisfying

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad (164)$$

$$\nabla \cdot u = 0, \quad u(0) = u_0, \quad (165)$$

$$p(t) = R_i R_j (u_i(t) u_j(t)), \quad \lim_{|x| \rightarrow \infty} p(t, x) = 0, \quad (166)$$

$$\frac{1}{2} \|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds = \frac{1}{2} \|u_0\|_{L^2}^2, \quad t \geq 0. \quad (167)$$

In particular,

$$\sup_{t \geq 0} \|u(t)\|_{L^2(\mathbb{R}^3)}^2 \leq \|u_0\|_{L^2(\mathbb{R}^3)}^2 < \infty. \quad (168)$$

This pair is unique among pressure-normalized solutions whose velocity belongs to $C([0, T]; H_\sigma^3(\mathbb{R}^3))$ for every finite $T > 0$.

Proof. Let

$$\mathcal{T} := \left\{ \tau > 0 : \begin{array}{l} \text{there exists a pressure-complete classical history} \\ (\tau, u^\tau, p^\tau) \text{ generated by } (u_0, \nu) \end{array} \right\},$$

and put

$$T_* := \sup \mathcal{T}.$$

Theorem 1.10 shows that $\mathcal{T}$ is nonempty. Strong uniqueness identifies any two such solutions on their common interval; their union defines a unique solution on $[0, T_*)$. Since $u_0 \in \mathcal{S}_\sigma$, the persistence conclusion of Theorem 1.10, applied on overlapping local intervals, places every strict restriction in the class of Definition 2.2. An extension beyond T_* would restrict to an interval $[0, \tau]$ with $\tau > T_*$, contrary to the definition of the supremum. Thus (T_*, u, p) is maximal. Corollary 4.7 gives $T_* = \infty$.

Fix $T < \infty$. Persistence in Theorem 1.10, iterated along the unique history, retains every finite spatial Sobolev order on $[0, T]$. Proposition 2.4 retains every finite temporal row. Lemma 1.2 gives (163); arbitrariness of T gives the global class. The pressure belongs to $H^s(\mathbb{R}^3)$ for every s , and the Fourier inversion proof of Lemma 1.2 places its continuous representative in $C_0(\mathbb{R}^3)$, which proves the limit in (166). Any second pressure-normalized solution in the stated uniqueness class agrees with u on each finite interval by the full $C([0, T]; H_\sigma^3)$ uniqueness in Theorem 1.10; pressure normalization then identifies its pressure with p . This proves global uniqueness in the asserted class.

Proposition 3.1 holds on every finite interval. Since T was arbitrary, it gives (167) for every $t \geq 0$; (168) follows by discarding the nonnegative dissipation term. $\square$

5 Quantitative consequences of the compatible rectangles

Definition 5.1 (Producer constants). Let $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, $\nu > 0$, and let (u, p) be the datum-generated pressure-normalized solution. Let $0 < T < \infty$ satisfy either $T < T_*$ or $T = T_* < \infty$, and set $I_T = (0, T)$,

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p, \quad n \in \mathbb{N}_0.$$

Define the monotone producer constant

$$\mathfrak{P}_{N,M}[u_0, \nu; T] := 1 + \max_{\substack{0 \leq N' \leq N \\ 0 \leq M' \leq M}} \sup_{0 < S < T} \mathfrak{R}_{N',M'}(u; S), \quad (169)$$

and the trace constant

$$\mathfrak{T}_{N,M}[u_0, \nu; T] := (1 + T^{-1})^{1/2} \mathfrak{P}_{N,M}[u_0, \nu; T]^{1/2}. \quad (170)$$

All occurrences of $\mathfrak{P}_{N,M}$ and $\mathfrak{T}_{N,M}$ below carry the arguments $[u_0, \nu; T]$.

Lemma 5.2 (Producer finiteness). *For every $N, M \in \mathbb{N}_0$,*

$$1 \leq \mathfrak{P}_{N,M}[u_0, \nu; T] < \infty. \quad (171)$$

Proof. If $T = T_* < \infty$, equation (110) proves (171). If $T < T_*$, classical persistence on $[0, T]$ gives

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} < \infty \quad (n, m \in \mathbb{N}_0),$$

and (72) proves (171). □

Lemma 5.3 (Fixed analytic constants). *For $r \in \mathbb{N}_0$, set*

$$A_r := 2C_r^{\text{alg}}.$$

Then

$$\|R_i R_j(f_i g_j)\|_{H^r} + \|\mathbb{P} \nabla \cdot (f \otimes g)\|_{H^{r-1}} \leq A_r \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}. \quad (172)$$

The same constant satisfies

$$\|R_i R_j(f_i g_j)\|_{H^r} + \|\nabla R_i R_j(f_i g_j)\|_{H^{r-1}} \leq A_r \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}. \quad (173)$$

For $s, k \in \mathbb{N}_0$, $2 \leq b \leq \infty$, and $s > k + \frac{3}{2} - \frac{3}{b}$, define

$$d_k := \#\{\alpha \in \mathbb{N}_0^3 : |\alpha| = k\}, \quad \frac{1}{q_b} := \frac{1}{2} - \frac{1}{b},$$

with $q_2 = \infty$, and

$$S_{s,k,b} := d_k^{1/2} (2\pi)^{-3(1/2-1/b)} \left\| |\xi|^k \langle \xi \rangle^{-s} \right\|_{L_\xi^{q_b}}. \quad (174)$$

Then $S_{s,k,b} < \infty$ and

$$\left\| \nabla^k f \right\|_{L^b} \leq S_{s,k,b} \|f\|_{H^s}. \quad (175)$$

For $m \geq 2$, set $L_m := \max\{1, C_m^{\text{alg}}\}$. Then

$$\|fg\|_{H^m} \leq L_m \|f\|_{H^m} \|g\|_{H^m}, \quad (176)$$

$$\|e^{\nu t \Delta} \mathbb{P} \nabla \cdot F\|_{H^m} \leq L_m (\nu t)^{-1/2} \|F\|_{H^m}, \quad t > 0. \quad (177)$$

Proof. Lemmas 1.4 and 1.5, applied componentwise before summing squares, give

$$\begin{aligned}\|R_i R_j(f_i g_j)\|_{H^r} &\leq \|f \otimes g\|_{H^r} \leq C_r^{\text{alg}} \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}, \\ \|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^{r-1}} &\leq \|f \otimes g\|_{H^r} \leq C_r^{\text{alg}} \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}.\end{aligned}$$

Moreover,

$$\|\nabla R_i R_j(f_i g_j)\|_{H^{r-1}} \leq \|R_i R_j(f_i g_j)\|_{H^r}.$$

This proves (172) and (173).

Let $b' = b/(b-1)$. Interpolation between Fourier inversion $L_\xi^1 \rightarrow L_x^\infty$, with norm $(2\pi)^{-3/2}$, and Plancherel gives

$$\|\mathcal{F}^{-1}h\|_{L_x^b} \leq (2\pi)^{-3(1/2-1/b)} \|h\|_{L_\xi^{b'}}.$$

For $|\alpha| = k$, Hölder's inequality with $1/b' = 1/2 + 1/q_b$ gives

$$\|\partial^\alpha f\|_{L^b} \leq (2\pi)^{-3(1/2-1/b)} \left\| |\xi|^k \langle \xi \rangle^{-s} \right\|_{L^{q_b}} \|f\|_{H^s}.$$

The weight belongs to L^{q_b} under the displayed strict inequality for s ; for $b = 2$, its L^∞ -norm is finite because $s > k$. Summation over $|\alpha| = k$ proves (175).

The first estimate in (176) is Lemma 1.4. For the heat estimate, the symbol of $\mathbb{P}\nabla \cdot$, acting from tensors to vectors, has norm at most $|\xi|$, while

$$\sup_{\rho \geq 0} \rho e^{-\nu t \rho^2} = (2e\nu t)^{-1/2}.$$

Plancherel gives

$$\|e^{\nu t \Delta} \mathbb{P}\nabla \cdot F\|_{H^m} \leq (2e\nu t)^{-1/2} \|F\|_{H^m} \leq L_m(\nu t)^{-1/2} \|F\|_{H^m}.$$

□

Definition 5.4 (Canonical rectangle majorants). For $N, M \in \mathbb{N}_0$, define

$$\mathfrak{B}_{N,M}[u_0, \nu; T] := \inf \{B \geq 0 : \mathfrak{R}_{N,M}(u; S) \leq B \text{ for every } 0 < S < T\}. \quad (178)$$

Monotone convergence gives

$$\mathfrak{B}_{N,M}[u_0, \nu; T] = \begin{cases} \mathfrak{R}_{N,M}(u; T), & T < T_*, \\ \mathfrak{R}_{N,M}^*[u_0, \nu, T_*], & T = T_* < \infty. \end{cases} \quad (179)$$

Equations (76) and (178) give the exact matrix identities

$$\mathfrak{B}_{N,M}[u_0, \nu; T] = \sup_{0 < S < T} \|\mathbf{G}_{N,M}(S)\|_{\mathbf{F}}^2, \quad (180)$$

$$\mathfrak{P}_{N,M}[u_0, \nu; T] = 1 + \max_{\substack{0 \leq N' \leq N \\ 0 \leq M' \leq M}} \sup_{0 < S < T} \|\mathbf{G}_{N',M'}(S)\|_{\mathbf{F}}^2. \quad (181)$$

Consequently,

$$\mathfrak{P}_{N,M}[u_0, \nu; T] = 1 + \max_{\substack{0 \leq N' \leq N \\ 0 \leq M' \leq M}} \mathfrak{B}_{N',M'}[u_0, \nu; T]. \quad (182)$$

For $n, m \in \mathbb{N}_0$, define the initial-jet majorants $\mathfrak{D}_{n,m} = \mathfrak{D}_{n,m}[u_0, \nu]$ by

$$\mathfrak{D}_{0,m} := \|u_0\|_{H^m}, \quad (183)$$

$$\mathfrak{D}_{n+1,m} := \nu \mathfrak{D}_{n,m+2} + A_{m+1} \sum_{a=0}^n \binom{n}{a} \mathfrak{D}_{a,m+2} \mathfrak{D}_{n-a,m+2}. \quad (184)$$

Define the pointwise recursion $\mathfrak{Y}_{n,m} = \mathfrak{Y}_{n,m}[u_0, \nu; T]$ by

$$\mathfrak{Y}_{0,0} := \|u_0\|_2, \quad (185)$$

$$\mathfrak{Y}_{0,m} := \min \left\{ (1 + T^{-1})^{1/2} \mathfrak{B}_{0,m}^{1/2}, \|u_0\|_{H^m} + T^{1/2} \mathfrak{B}_{0,m+1}^{1/2} \right\}, \quad m \geq 1, \quad (186)$$

$$\mathfrak{Y}_{n+1,m} := \nu \mathfrak{Y}_{n,m+2} + A_{m+1} \sum_{a=0}^n \binom{n}{a} \mathfrak{Y}_{a,m+2} \mathfrak{Y}_{n-a,m+2}. \quad (187)$$

For $m \in \{-1\} \cup \mathbb{N}_0$, put

$$m_+ := \max\{m, 0\}, \quad \mu(m) := \max\{m - 1, 0\},$$

and define

$$\tilde{\mathfrak{X}}_{n,m} := \min \left\{ T^{1/2} \mathfrak{Y}_{n,m_+}, \mathfrak{B}_{n,\mu(m)}^{1/2} \right\}. \quad (188)$$

Set

$$\mathfrak{X}_{0,0} := \min \left\{ \tilde{\mathfrak{X}}_{0,0}, T^{1/2} \|u_0\|_2 \right\}, \quad (189)$$

$$\mathfrak{X}_{0,1} := \min \left\{ \tilde{\mathfrak{X}}_{0,1}, \left(T + \frac{1}{2\nu} \right)^{1/2} \|u_0\|_2 \right\}, \quad (190)$$

$$\mathfrak{X}_{n,m} := \tilde{\mathfrak{X}}_{n,m} \quad \text{for all other } (n, m). \quad (191)$$

For $n, m \in \mathbb{N}_0$, define

$$\mathfrak{Z}_{n,m}[u_0, \nu; T] := \min \left\{ \mathfrak{Y}_{n,m}, (1 + T^{-1})^{1/2} \mathfrak{B}_{n,m}^{1/2}, \mathfrak{D}_{n,m} + T^{1/2} \mathfrak{X}_{n+1,m} \right\}. \quad (192)$$

All $\mathfrak{B}, \mathfrak{D}, \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z}$ in this section carry the arguments $[u_0, \nu; T]$, except that $\mathfrak{D}$ is independent of T .

Lemma 5.5 (Closed velocity majorants). *For every $n, m \in \mathbb{N}_0$,*

$$\|V_n(0)\|_{H^m} \leq \mathfrak{D}_{n,m}, \quad (193)$$

$$\|V_n\|_{L^2(I_T; H^m)} \leq \mathfrak{X}_{n,m}, \quad (194)$$

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} \leq \mathfrak{Z}_{n,m}. \quad (195)$$

For $m = -1$, equation (194) remains valid. Every right-hand side is finite and depends on (u_0, ν, n, m, T) through the finite producer matrix suprema in (180).

Proof. Applying $\mathbb{P}$ to the temporal recurrence gives

$$V_{n+1} = \nu \Delta V_n - \mathbb{P} \nabla \cdot \sum_{a=0}^n \binom{n}{a} (V_a \otimes V_{n-a}). \quad (196)$$

Equation (172) at order $m + 1$, evaluated at $t = 0$, proves (193) by induction through (184). The same estimate, taken pointwise on $[0, T]$, proves by induction that

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} \leq \mathfrak{Y}_{n,m}. \quad (197)$$

The base case $m = 0$ is (52). For $m \geq 1$, both terms in (186) are valid: the first is the adjacent trace bound, and the second follows from

$$u(t) = u_0 + \int_0^t V_1(s) \, ds \quad \text{in } H^m.$$

The rectangle with middle order $\mu(m)$, Sobolev nesting, and (197) give respectively

$$\|V_n\|_{L^2(I_T; H^m)} \leq \mathfrak{B}_{n, \mu(m)}^{1/2}, \quad \|V_n\|_{L^2(I_T; H^m)} \leq T^{1/2} \mathfrak{Y}_{n, m+}.$$

This proves (194) through (188); the two exceptional improvements are (52) and (53). The adjacent trace lemma gives

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} \leq (1 + T^{-1})^{1/2} \mathfrak{B}_{n,m}^{1/2},$$

while the Banach-valued fundamental identity and (193) give

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} \leq \mathfrak{D}_{n,m} + T^{1/2} \mathfrak{X}_{n+1,m}.$$

Taking the minimum of these bounds and (197) proves (195). $\square$

Definition 5.6 (Closed pressure majorants). For $n, m \in \mathbb{N}_0$, define

$$\mathfrak{Q}_{n,m}^{(1)} := A_m \sum_{a=0}^n \binom{n}{a} \mathfrak{X}_{a,m+1} \mathfrak{X}_{n-a,m+1}, \quad (198)$$

$$\mathfrak{Q}_{n,m}^{(2)} := A_m \sum_{a=0}^n \binom{n}{a} \min \{ \mathfrak{Z}_{a,m+1} \mathfrak{X}_{n-a,m+1}, \mathfrak{X}_{a,m+1} \mathfrak{Z}_{n-a,m+1} \}, \quad (199)$$

$$\mathfrak{Q}_{n,m}^{(\infty)} := A_m \sum_{a=0}^n \binom{n}{a} \mathfrak{Z}_{a,m+1} \mathfrak{Z}_{n-a,m+1}, \quad (200)$$

$$\mathfrak{Q}_{n,m}^{(\text{mod})} := A_m \sum_{a=0}^n \binom{n}{a} (\mathfrak{X}_{a+1,m+1} \mathfrak{Z}_{n-a,m+1} + \mathfrak{Z}_{a,m+1} \mathfrak{X}_{n-a+1,m+1}). \quad (201)$$

These quantities carry the arguments $[u_0, \nu; T]$ through $\mathfrak{X}$ and $\mathfrak{Z}$.

Theorem 5.7 (Quantitative mixed-jet rows and traces). *Let $N, s \in \mathbb{N}_0$, let $0 \leq n \leq N$, and let $\alpha \in \mathbb{N}_0^3$. Put*

$$J_{n,\alpha} := \partial_x^\alpha V_n.$$

Then

$$\|J_{n,\alpha}\|_{L^2(I_T; H^{s+1})}^2 + \|J_{n+1,\alpha}\|_{L^2(I_T; H^{s-1})}^2 \leq \mathfrak{P}_{N,s+|\alpha|}, \quad (202)$$

$$\sup_{0 \leq t \leq T} \|J_{n,\alpha}(t)\|_{H^s} \leq \mathfrak{T}_{N,s+|\alpha|}. \quad (203)$$

There is a unique trace $J_{n,\alpha}^T \in H^s$ at $t = T$, and for $0 \leq r \leq t \leq T$,

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^{s-1}} \leq (t-r)^{1/2} \mathfrak{P}_{N,s+|\alpha|}^{1/2}, \quad (204)$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^s} \leq (t-r)^{1/2} \mathfrak{P}_{N,s+|\alpha|+1}^{1/2}, \quad (205)$$

$$\|J_{n,\alpha}^T\|_{H^s} \leq \mathfrak{T}_{N,s+|\alpha|}. \quad (206)$$

If $0 \leq \theta \leq 1$, then

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^{s-\theta}} \leq 2^{1-\theta} \mathfrak{T}_{N,s+|\alpha|}^{1-\theta} \mathfrak{P}_{N,s+|\alpha|}^{\theta/2} (t-r)^{\theta/2}. \quad (207)$$

For $1 \leq a \leq \infty$, $2 \leq b \leq \infty$, $k, q \in \mathbb{N}_0$, and $q > k + \frac{3}{2} - \frac{3}{b}$,

$$\left\| \nabla^k J_{n,\alpha} \right\|_{L^a(I_T; L^b)} \leq S_{q,k,b} T^{1/a} \mathfrak{T}_{N,q+|\alpha|}, \quad (208)$$

$$\left\| \nabla^k J_{n,\alpha} \right\|_{L^2(I_T; L^b)} \leq S_{q,k,b} \mathfrak{P}_{N,q+|\alpha|}^{1/2}, \quad (209)$$

where $T^{1/\infty} = 1$.

Proof. Take the rectangle with middle order $M = s + |\alpha|$. The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^r} \leq \|f\|_{H^{r+|\alpha|}}$$

and (169) give (202). The adjacent-row trace estimate gives

$$\begin{aligned} \sup_{0 \leq t \leq T} \|J_{n,\alpha}(t)\|_{H^s}^2 &\leq T^{-1} \|J_{n,\alpha}\|_{L^2(I_T; H^s)}^2 \\ &\quad + 2 \|J_{n,\alpha}\|_{L^2(I_T; H^{s+1})} \|J_{n+1,\alpha}\|_{L^2(I_T; H^{s-1})} \\ &\leq (T^{-1} + 1) \mathfrak{P}_{N,s+|\alpha|}, \end{aligned}$$

which proves (203) and (206). The Bochner identity

$$J_{n,\alpha}(t) - J_{n,\alpha}(r) = \int_r^t J_{n+1,\alpha}(\tau) d\tau$$

in H^{s-1} , followed by Cauchy-Schwarz, proves (204). Applying the same argument to the rectangle with middle order $s + |\alpha| + 1$ proves (205). Interpolation between H^{s-1} and H^s , together with $\|J(t) - J(r)\|_{H^s} \leq 2\mathfrak{T}_{N,s+|\alpha|}$, proves (207). Sobolev embedding and (203) prove (208); Sobolev embedding and (202) prove (209). $\square$

Theorem 5.8 (Quantitative pressure rows). *Let $N, s \in \mathbb{N}_0$, $0 \leq n \leq N$, and $\alpha \in \mathbb{N}_0^3$. Set*

$$\Pi_{n,\alpha} := \partial_x^\alpha Q_n, \quad r := s + |\alpha|.$$

Then

$$\|\Pi_{n,\alpha}\|_{L^1(I_T; H^s)} + \|\nabla \Pi_{n,\alpha}\|_{L^1(I_T; H^{s-1})} \leq 2A_r 2^n \mathfrak{P}_{N,r}, \quad (210)$$

$$\|\Pi_{n,\alpha}\|_{L^2(I_T; H^s)} \leq A_r 2^n \mathfrak{T}_{N,r+1} \mathfrak{P}_{N,r}^{1/2}, \quad (211)$$

$$\sup_{0 \leq t \leq T} \|\Pi_{n,\alpha}(t)\|_{H^s} \leq A_r 2^n \mathfrak{T}_{N,r+1}^2. \quad (212)$$

There is a unique endpoint pressure row $\Pi_{n,\alpha}^T \in H^s$, and

$$\|\Pi_{n,\alpha}^T\|_{H^s} \leq A_r 2^n \mathfrak{T}_{N,r+1}^2, \quad (213)$$

$$\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}(r_0)\|_{H^s} \leq 2^{n+1} A_r \mathfrak{T}_{N,r+1} \mathfrak{P}_{N,r+2}^{1/2} |t - r_0|^{1/2} \quad (214)$$

for $0 \leq r_0 \leq t \leq T$. If $1 \leq a \leq \infty$, $2 \leq b \leq \infty$, $k, q \in \mathbb{N}_0$, and $q > k + \frac{3}{2} - \frac{3}{b}$, then

$$\left\| \nabla^k \Pi_{n,\alpha} \right\|_{L^a(I_T; L^b)} \leq S_{q,k,b} T^{1/a} A_{q+|\alpha|} 2^n \mathfrak{T}_{N,q+|\alpha|+1}^2. \quad (215)$$

Proof. The normalized pressure recurrence is

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}. \quad (216)$$

Equation (172), Cauchy–Schwarz in time, and $\sum_{a=0}^n \binom{n}{a} = 2^n$ give

$$\|Q_n\|_{L^1(I_T; H^r)} \leq A_r 2^n \mathfrak{P}_{N,r}. \quad (217)$$

The gradient estimate in (210) follows from $\|\nabla f\|_{H^{r-1}} \leq \|f\|_{H^r}$. Using one factor in $L_t^\infty H_x^{r+1}$ and the other in $L_t^2 H_x^{r+1}$ proves (211). Using both factors in $L_t^\infty H_x^{r+1}$ proves (212) and (213).

Subtract (216) at times t and r_0 , split each product difference into two terms, apply (205) in H^{r+1} , and use (203). The two product differences and the binomial sum give (214). Equation (215) follows from (212) and Sobolev embedding. $\square$

Proposition 5.9 (Matrix form of the closed rectangle estimates). *For nonnegative matrices of the same size, write $A \preceq_{\text{ent}} B$ when every entry of A is at most the corresponding entry of B . When $T = T_* < \infty$, interpret the closed rectangle matrix entrywise by*

$$\mathbf{C}_{N,M}(T_*) := \lim_{S \uparrow T_*} \mathbf{C}_{N,M}(S).$$

The limit exists in $[0, \infty]^{(N+1) \times 3}$ by monotonicity of every column norm, and the estimates below show that all of its entries are finite. Then

$$\mathbf{C}_{N,M}(T) \preceq_{\text{ent}} \left(\begin{array}{ccc} \mathfrak{X}_{n,M+1} & \mathfrak{X}_{n+1,M-1} & \mathfrak{Q}_{n,M}^{(1)} \end{array} \right)_{n=0}^N. \quad (218)$$

Consequently,

$$\mathfrak{R}_{N,M}(u; T) \leq \sum_{n=0}^N (\mathfrak{X}_{n,M+1}^2 + \mathfrak{X}_{n+1,M-1}^2), \quad (219)$$

$$\mathfrak{Q}_{N,M}(p; T) \leq \sum_{n=0}^N \mathfrak{Q}_{n,M}^{(1)}. \quad (220)$$

Proof. The first two columns of (218) are (194). The pressure recurrence (216), the product estimate (173), and Cauchy–Schwarz in time give the third column through (198). Take the Frobenius norm of the first two columns and sum the third column, using (76). $\square$

Proposition 5.10 (Velocity and pressure frequency tails). *Let $P_{\geq K}$ be the Fourier multiplier of $\mathbf{1}_{\{|\xi| \geq K\}}$, where $K \geq 1$. Let $r_1 \in \mathbb{N}$, $s \in \mathbb{N}_0$, $0 \leq n \leq N$, and $\alpha \in \mathbb{N}_0^3$. Then*

$$\|P_{\geq K} J_{n,\alpha}\|_{L^2(I_T; H^s)} \leq K^{-r_1} \mathfrak{P}_{N,s+r_1+|\alpha|}^{1/2}, \quad (221)$$

$$\sup_{0 \leq t \leq T} \|P_{\geq K} J_{n,\alpha}(t)\|_{H^s} \leq K^{-r_1} \mathfrak{T}_{N,s+r_1+|\alpha|}, \quad (222)$$

$$\|P_{\geq K} \Pi_{n,\alpha}\|_{L^1(I_T; H^s)} \leq K^{-r_1} A_{s+r_1+|\alpha|} 2^n \mathfrak{P}_{N,s+r_1+|\alpha|}, \quad (223)$$

$$\sup_{0 \leq t \leq T} \|P_{\geq K} \Pi_{n,\alpha}(t)\|_{H^s} \leq K^{-r_1} A_{s+r_1+|\alpha|} 2^n \mathfrak{T}_{N,s+r_1+|\alpha|+1}^2. \quad (224)$$

The pointwise estimates (222) and (224) hold at $t = T$.

Proof. For every $f \in H^{s+r_1}$,

$$\|P_{\geq K} f\|_{H^s} \leq K^{-r_1} \|f\|_{H^{s+r_1}}.$$

Apply this inequality to (202) and (203) at spatial order $s + r_1$ to obtain the two velocity bounds. Since Fourier multipliers commute,

$$\|P_{\geq K} \Pi_{n,\alpha}\|_{H^s} \leq K^{-r_1} \|Q_n\|_{H^{s+r_1+|\alpha|}}.$$

Integration in time followed by (217) at order $s + r_1 + |\alpha|$ gives (223). Applying the tail inequality directly to $\Pi_{n,\alpha}$, followed by (212) with s replaced by $s + r_1$, gives (224). $\square$

Proposition 5.11 (Critical-height bounds). *Define*

$$E_\sigma(t) := \frac{1}{2} \|\Lambda^\sigma u(t)\|_2^2, \quad H(t) := E_{1/2}(t),$$

and

$$E_{j+1/2, \geq K}(t) := \frac{1}{2} \int_{|\xi| \geq K} |\xi|^{2j+1} |\widehat{u}(t, \xi)|^2 d\xi.$$

For $j \in \mathbb{N}_0$,

$$\sup_{0 \leq t \leq T} E_{j+1/2}(t) \leq \frac{1}{2} \mathfrak{T}_{0,j+1}^2, \quad (225)$$

$$\int_0^T E_{j+1/2}(t) dt \leq \frac{1}{2} \mathfrak{P}_{0,j}. \quad (226)$$

If $m > j + \frac{1}{2}$ is an integer and $K \geq 1$, then

$$\sup_{0 \leq t \leq T} E_{j+1/2, \geq K}(t) \leq \frac{1}{2} K^{2j+1-2m} \mathfrak{T}_{0,m}^2, \quad (227)$$

$$\int_0^T E_{j+1/2, \geq K}(t) dt \leq \frac{1}{2} K^{2j+1-2m} \mathfrak{P}_{0,m-1}. \quad (228)$$

For $k \in \mathbb{N}$,

$$\|H^{(k)}\|_{L^1(0,T)} \leq 2^{k-1} \mathfrak{P}_{k,0}. \quad (229)$$

In particular,

$$\|H'\|_{L^1(0,T)} \leq \frac{1}{2} \mathfrak{P}_{0,2}, \quad (230)$$

$$\int_0^T \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt \leq \mathfrak{P}_{0,1}, \quad (231)$$

$$\left\| \mathcal{P}_{1/2} \right\|_{L^1(0,T)} \leq \frac{1}{2} \mathfrak{P}_{0,2} + \nu \mathfrak{P}_{0,1}, \quad (232)$$

$$|H(T) - H(t)| \leq \mathfrak{T}_{0,1} \mathfrak{P}_{0,2}^{1/2} (T-t)^{1/2}, \quad 0 \leq t \leq T, \quad (233)$$

where $\mathcal{P}_{1/2} = H' + \nu \left\| \Lambda^{3/2} u \right\|_2^2$.

Proof. Equations (225) and (226) follow from $\dot{H}^{j+1/2} \leftrightarrow H^{j+1}$, (203), and (202). On $|\xi| \geq K$,

$$|\xi|^{2j+1} \leq K^{2j+1-2m} |\xi|^{2m},$$

which proves (227) and (228).

The identity

$$H^{(k)}(t) = \frac{1}{2} \sum_{a=0}^k \binom{k}{a} \operatorname{Re} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{k-a}(t) \right\rangle$$

and Cauchy–Schwarz in time give (229). At $k = 1$, the two equal terms give

$$\left\| H' \right\|_{L^1(0,T)} \leq \|u_t\|_{L^2(I_T; \dot{H}^{1/2})} \|u\|_{L^2(I_T; \dot{H}^{1/2})} \leq \frac{1}{2} \mathfrak{P}_{0,2}.$$

Equation (231) follows from the upper velocity row of the $(0, 1)$ rectangle. The identity defining $\mathcal{P}_{1/2}$ gives (232). Finally,

$$\|u(T) - u(t)\|_{\dot{H}^{1/2}} \leq \|u(T) - u(t)\|_{H^1} \leq (T-t)^{1/2} \|u_t\|_{L^2(t,T; H^1)} \leq (T-t)^{1/2} \mathfrak{P}_{0,2}^{1/2},$$

and

$$|H(T) - H(t)| \leq \frac{1}{2} (\|u(T)\|_{\dot{H}^{1/2}} + \|u(t)\|_{\dot{H}^{1/2}}) \|u(T) - u(t)\|_{\dot{H}^{1/2}}.$$

Equation (203) bounds each retained velocity factor, so these two estimates prove (233). $\square$

Proposition 5.12 (Continuation norms). *The finite-horizon solution satisfies*

$$\|u\|_{L^2(I_T; L^\infty)} \leq S_{2,0,\infty} \mathfrak{P}_{0,1}^{1/2}, \quad (234)$$

$$\int_0^T \|\nabla u(t)\|_{L^\infty} dt \leq S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}^{1/2}, \quad (235)$$

$$\int_0^T \|\nabla \times u(t)\|_{L^\infty} dt \leq 2S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}^{1/2}. \quad (236)$$

For every integer $m \geq 3$, set $K_m := C_m^{\text{com}}$. Then

$$|\operatorname{Re} \langle A^m \mathbb{P} \nabla \cdot (u \otimes u), A^m u \rangle_{L^2}| \leq K_m \|\nabla u\|_{L^\infty} \|u\|_{H^m}^2. \quad (237)$$

$$\sup_{0 \leq t \leq T} \|u(t)\|_{H^m}^2 \leq \|u_0\|_{H^m}^2 \exp \left(2K_m S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}^{1/2} \right). \quad (238)$$

Proof. Sobolev embedding and the upper rows of the $(0, 1)$ and $(0, 2)$ rectangles give (234) and (235); the pointwise bound $\|\nabla \times u\|_\infty \leq 2 \|\nabla u\|_\infty$ gives (236). The H^m energy identity and (237) give

$$\frac{1}{2} \frac{d}{dt} \|u\|_{H^m}^2 + \nu \|\nabla u\|_{H^m}^2 \leq K_m \|\nabla u\|_{L^\infty} \|u\|_{H^m}^2,$$

and hence

$$\frac{d}{dt} \|u\|_{H^m}^2 \leq 2K_m \|\nabla u\|_{L^\infty} \|u\|_{H^m}^2.$$

Gronwall's inequality and (235) prove (238). $\square$

Theorem 5.13 (Quantitative endpoint restart). *Assume $T = T_* < \infty$, and set*

$$u_T := u_* := V_0^*, \quad V_n^T := V_n^*, \quad Q_n^T := Q_n^*.$$

Define

$$\mathcal{L}_T := \left\{ \delta > 0 : \begin{array}{l} \exists (z, \pi) \text{ a pressure-normalized solution on } [T, T + \delta], \\ z(T) = u_T, \quad z \in C([T, T + \delta]; H_\sigma^3) \end{array} \right\}, \quad (239)$$

$$\delta_T^{\max} := \sup \mathcal{L}_T.$$

Define

$$\delta_{\text{qc}}[u_0, \nu; T] := \min \left\{ 1, c_\nu (1 + \mathfrak{Z}_{0,3}[u_0, \nu; T])^{-2} \right\}. \quad (240)$$

The pressure-normalized solution (w, q) with $w(T) = u_T$ exists uniquely on $[T, T + \delta_{\text{qc}}]$, and, for every integer $m \geq 3$,

$$w \in C([T, T + \delta_{\text{qc}}]; H_\sigma^m) \cap L^2(T, T + \delta_{\text{qc}}; H^{m+1}), \quad w_t \in L^2(T, T + \delta_{\text{qc}}; H^{m-1}). \quad (241)$$

At the base restart order,

$$\begin{aligned} & \|w\|_{C([T, T + \delta_{\text{qc}}]; H^3)} + \|w\|_{L^2(T, T + \delta_{\text{qc}}; H^4)} + \|w_t\|_{L^2(T, T + \delta_{\text{qc}}; H^2)} \\ & \leq 2L_\nu \mathfrak{Z}_{0,3}[u_0, \nu; T]. \end{aligned} \quad (242)$$

Set

$$\mathfrak{A}_T := 2S_{3,1,\infty} L_\nu \delta_{\text{qc}} \mathfrak{Z}_{0,3}[u_0, \nu; T]. \quad (243)$$

For every integer $m \geq 3$,

$$\sup_{T \leq t \leq T + \delta_{\text{qc}}} \|w(t)\|_{H^m}^2 \leq \mathfrak{Z}_{0,m}[u_0, \nu; T]^2 \exp(2K_m \mathfrak{A}_T). \quad (244)$$

Moreover, $w, q \in C^\infty([T, T + \delta_{\text{qc}}] \times \mathbb{R}^3)$. Define

$$W_n^T := \partial_t^n w(T), \quad P_n^T := \partial_t^n q(T).$$

Then

$$W_n^T = V_n^T, \quad P_n^T = Q_n^T, \quad n \in \mathbb{N}_0. \quad (245)$$

The maximal restart lifespan $\delta_T^{\max}$ obeys

$$\delta_T^{\max} \geq \delta_{\text{qc}}[u_0, \nu; T] > 0. \quad (246)$$

Proof. Equations (141) and (195) give

$$\|u_T\|_{H^3} \leq \mathfrak{Z}_{0,3}.$$

Consequently,

$$\delta_{\text{qc}}[u_0, \nu; T] \leq \min\{1, c_\nu (1 + \|u_T\|_{H^3})^{-2}\}.$$

Theorem 1.10, translated to time T , gives a solution on the interval in the statement. The fixed-point ball in its proof gives (242). Hence

$$\int_T^{T + \delta_{\text{qc}}} \|\nabla w(t)\|_\infty \, dt \leq S_{3,1,\infty} \delta_{\text{qc}} \|w\|_{C([T, T + \delta_{\text{qc}}]; H^3)} \leq \mathfrak{A}_T.$$

The H^m energy estimate in Proposition 5.12, now on the restart interval, and $\|u_T\|_{H^m} \leq \mathfrak{Z}_{0,m}$ prove (244). Equation (145) gives $u_T \in H^\infty_\sigma$. The persistence assertion in Theorem 1.10 proves (241) for every $m \geq 3$ and gives the stated smoothness before the endpoint rows are defined. The full-class uniqueness proved in Theorem 1.10 gives the uniqueness asserted here.

The endpoint jet from Theorem 4.4 satisfies

$$Q_n^T = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}^T V_{n-a,j}^T,$$

$$V_{n+1}^T = \nu \Delta V_n^T - \sum_{a=0}^n \binom{n}{a} (V_a^T \cdot \nabla) V_{n-a}^T - \nabla Q_n^T.$$

The smooth right-hand solution satisfies the same two recurrences with (W_n^T, P_n^T) . Since $W_0^T = u_T = V_0^T$, induction in n proves (245). Theorem 4.6 and strong uniqueness identify this solution with the continuation of the same preterminal history. Its existence on the displayed interval proves (246). $\square$

Theorem 5.14 (Sharp datum-indexed finite-horizon estimates). *Let $n, s, k, q \in \mathbb{N}_0$, $\alpha \in \mathbb{N}_0^3$, and set $r := s + |\alpha|$. Then*

$$\|J_{n,\alpha}\|_{L^2(I_T; H^{s+1})} \leq \mathfrak{X}_{n,r+1}, \quad (247)$$

$$\|J_{n+1,\alpha}\|_{L^2(I_T; H^{s-1})} \leq \mathfrak{X}_{n+1,r-1}, \quad (248)$$

$$\sup_{0 \leq t \leq T} \|J_{n,\alpha}(t)\|_{H^s} \leq \mathfrak{Z}_{n,r}, \quad (249)$$

$$\|J_{n,\alpha}^T\|_{H^s} \leq \mathfrak{Z}_{n,r}, \quad (250)$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^{s-1}} \leq \min \left\{ |t - r_0|^{1/2} \mathfrak{X}_{n+1,r-1}, |t - r_0| \mathfrak{Z}_{n+1,r} \right\}, \quad (251)$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^s} \leq \min \left\{ |t - r_0|^{1/2} \mathfrak{X}_{n+1,r}, |t - r_0| \mathfrak{Z}_{n+1,r} \right\}, \quad (252)$$

for $0 \leq r_0 \leq t \leq T$. For $0 \leq \theta \leq 1$,

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^{s-\theta}} \leq 2^{1-\theta} \mathfrak{Z}_{n,r}^{1-\theta} \mathfrak{X}_{n+1,r-1}^\theta |t - r_0|^{\theta/2}. \quad (253)$$

If $1 \leq a \leq \infty$, $2 \leq b \leq \infty$, and $q > k + \frac{3}{2} - \frac{3}{b}$, then

$$\left\| \nabla^k J_{n,\alpha} \right\|_{L^a(I_T; L^b)} \leq S_{q,k,b} T^{1/a} \mathfrak{Z}_{n,q+|\alpha|}, \quad (254)$$

$$\left\| \nabla^k J_{n,\alpha} \right\|_{L^2(I_T; L^b)} \leq S_{q,k,b} \mathfrak{X}_{n,q+|\alpha|}, \quad (255)$$

where $T^{1/\infty} = 1$.

The pressure rows obey

$$\|\Pi_{n,\alpha}\|_{L^1(I_T; H^s)} + \|\nabla \Pi_{n,\alpha}\|_{L^1(I_T; H^{s-1})} \leq \mathfrak{Q}_{n,r}^{(1)}, \quad (256)$$

$$\|\Pi_{n,\alpha}\|_{L^2(I_T; H^s)} \leq \mathfrak{Q}_{n,r}^{(2)}, \quad (257)$$

$$\sup_{0 \leq t \leq T} \|\Pi_{n,\alpha}(t)\|_{H^s} \leq \mathfrak{Q}_{n,r}^{(\infty)}, \quad (258)$$

$$\|\Pi_{n,\alpha}^T\|_{H^s} \leq \mathfrak{Q}_{n,r}^{(\infty)}, \quad (259)$$

$$\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}(r_0)\|_{H^s} \leq \min \left\{ |t - r_0|^{1/2} \mathfrak{Q}_{n,r}^{(\text{mod})}, |t - r_0| \mathfrak{Q}_{n+1,r}^{(\infty)} \right\}. \quad (260)$$

Under the preceding restrictions on a, b, k, q ,

$$\left\| \nabla^k \Pi_{n,\alpha} \right\|_{L^a(I_T; L^b)} \leq S_{q,k,b} T^{1/a} \mathfrak{Q}_{n,q+|\alpha|}^{(\infty)}. \quad (261)$$

Let $r_1 \in \mathbb{N}$ and $K \geq 1$. Then

$$\|\mathbf{P}_{\geq K} J_{n,\alpha}\|_{L^2(I_T; H^s)} \leq K^{-r_1} \mathfrak{X}_{n,s+r_1+|\alpha|}, \quad (262)$$

$$\sup_{0 \leq t \leq T} \|\mathbf{P}_{\geq K} J_{n,\alpha}(t)\|_{H^s} \leq K^{-r_1} \mathfrak{Z}_{n,s+r_1+|\alpha|}, \quad (263)$$

$$\|\mathbf{P}_{\geq K} \Pi_{n,\alpha}\|_{L^1(I_T; H^s)} \leq K^{-r_1} \mathfrak{Q}_{n,s+r_1+|\alpha|}^{(1)}, \quad (264)$$

$$\sup_{0 \leq t \leq T} \|\mathbf{P}_{\geq K} \Pi_{n,\alpha}(t)\|_{H^s} \leq K^{-r_1} \mathfrak{Q}_{n,s+r_1+|\alpha|}^{(\infty)}. \quad (265)$$

For $j \in \mathbb{N}_0$,

$$\sup_{0 \leq t \leq T} E_{j+1/2}(t) \leq \frac{1}{2} \mathfrak{Z}_{0,j+1}^2, \quad (266)$$

$$\int_0^T E_{j+1/2}(t) dt \leq \frac{1}{2} \mathfrak{X}_{0,j+1}^2. \quad (267)$$

If $m > j + \frac{1}{2}$ is an integer, then

$$\sup_{0 \leq t \leq T} E_{j+1/2, \geq K}(t) \leq \frac{1}{2} K^{2j+1-2m} \mathfrak{Z}_{0,m}^2, \quad (268)$$

$$\int_0^T E_{j+1/2, \geq K}(t) dt \leq \frac{1}{2} K^{2j+1-2m} \mathfrak{X}_{0,m}^2. \quad (269)$$

For $k \in \mathbb{N}$,

$$\left\| H^{(k)} \right\|_{L^1(0,T)} \leq \frac{1}{2} \sum_{a=0}^k \binom{k}{a} \mathfrak{X}_{a,1} \mathfrak{X}_{k-a,1}. \quad (270)$$

For $j, \ell \in \mathbb{N}_0$, define

$$\begin{aligned} \mathfrak{E}_{j,\ell}^{[1]} &:= \frac{1}{2} \sum_{a=0}^{\ell} \binom{\ell}{a} \mathfrak{X}_{a,j+1} \mathfrak{X}_{\ell-a,j+1}, \\ \mathfrak{E}_{j,\ell}^{[\infty]} &:= \frac{1}{2} \sum_{a=0}^{\ell} \binom{\ell}{a} \mathfrak{Z}_{a,j+1} \mathfrak{Z}_{\ell-a,j+1}, \end{aligned} \quad (271)$$

$$\begin{aligned} \mathfrak{E}_{j,\ell}^{[1]} &:= \mathfrak{E}_{j,\ell+1}^{[1]} + 2\nu \mathfrak{E}_{j+1,\ell}^{[1]}, \\ \mathfrak{E}_{j,\ell}^{[\infty]} &:= \mathfrak{E}_{j,\ell+1}^{[\infty]} + 2\nu \mathfrak{E}_{j+1,\ell}^{[\infty]}. \end{aligned} \quad (272)$$

Then the complete differentiated energy and production rows obey

$$\left\| \partial_t^\ell E_{j+1/2} \right\|_{L^1(0,T)} \leq \mathfrak{E}_{j,\ell}^{[1]}, \quad \sup_{0 \leq t \leq T} \left| \partial_t^\ell E_{j+1/2}(t) \right| \leq \mathfrak{E}_{j,\ell}^{[\infty]}, \quad (273)$$

$$\left\| \partial_t^\ell \mathcal{P}_{j+1/2} \right\|_{L^1(0,T)} \leq \mathfrak{E}_{j,\ell}^{[1]}, \quad \sup_{0 \leq t \leq T} \left| \partial_t^\ell \mathcal{P}_{j+1/2}(t) \right| \leq \mathfrak{E}_{j,\ell}^{[\infty]}. \quad (274)$$

Every current in (123) now has a finite-order majorant retaining temporal depth, spatial height, viscosity, and horizon. In particular,

$$\|H'\|_{L^1(0,T)} \leq \mathfrak{X}_{1,1}\mathfrak{X}_{0,1}, \quad (275)$$

$$\int_0^T \|\Lambda^{3/2}u(t)\|_2^2 dt \leq \mathfrak{X}_{0,2}^2, \quad (276)$$

$$\|\mathcal{P}_{1/2}\|_{L^1(0,T)} \leq \mathfrak{X}_{1,1}\mathfrak{X}_{0,1} + \nu\mathfrak{X}_{0,2}^2, \quad (277)$$

$$|H(T) - H(t)| \leq \min \left\{ \mathfrak{Z}_{0,1}\mathfrak{X}_{1,1}(T-t)^{1/2}, \mathfrak{Z}_{0,1}\mathfrak{Z}_{1,1}(T-t) \right\}, \quad 0 \leq t \leq T. \quad (278)$$

Finally,

$$\|u\|_{L^2(I_T;L^\infty)} \leq S_{2,0,\infty}\mathfrak{X}_{0,2}, \quad (279)$$

$$\int_0^T \|\nabla u(t)\|_\infty dt \leq S_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}, \quad (280)$$

$$\int_0^T \|\nabla \times u(t)\|_\infty dt \leq 2S_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}, \quad (281)$$

$$\sup_{0 \leq t \leq T} \|u(t)\|_{H^m}^2 \leq \|u_0\|_{H^m}^2 \exp\left(2K_m S_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}\right), \quad m \geq 3. \quad (282)$$

Every right-hand side is a finite expression in the displayed analytic constants, initial-jet norms, and the finite producer matrix suprema (180). An explicit datum-norm formula for the latter suprema propagates through all of these estimates by direct substitution.

Proof. The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^\ell} \leq \|f\|_{H^{\ell+|\alpha|}},$$

followed by Lemma 5.5, proves (247)–(250). The Banach identity

$$J_{n,\alpha}(t) - J_{n,\alpha}(r_0) = \int_{r_0}^t J_{n+1,\alpha}(\tau) d\tau$$

in the two indicated Sobolev spaces and Cauchy–Schwarz prove the square-root branches of (251) and (252). The same identity, followed by the pointwise bound (195) for V_{n+1} , proves their linear branches. Interpolation between the low modulus and

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^s} \leq 2\mathfrak{Z}_{n,r}$$

proves (253). The fixed Sobolev embedding constant in (175) proves (254) and (255).

Apply the recurrence

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}$$

term by term at order r , using (173). Cauchy–Schwarz in time gives (256); placing either factor in $L_t^\infty H_x^{r+1}$ gives (257); placing both factors there gives (258) and (259). After subtracting the recurrence at times t and r_0 , split each product difference into two terms and apply (252); this gives the square-root branch of (260). Since $\partial_t \Pi_{n,\alpha} = \Pi_{n+1,\alpha}$, the Banach identity and (258) at row $n+1$ give its linear branch. Sobolev embedding proves (261).

For every $f \in H^{\ell+r_1}$,

$$\|P_{\geq K} f\|_{H^\ell} \leq K^{-r_1} \|f\|_{H^{\ell+r_1}}.$$

Applying this inequality to the preceding velocity and pressure estimates proves (262)–(265).

The inequalities

$$\|f\|_{\dot{H}^{j+1/2}} \leq \|f\|_{H^{j+1}}, \quad |\xi|^{2j+1} \leq K^{2j+1-2m} |\xi|^{2m} \quad (|\xi| \geq K),$$

give (266)–(269). The exact identity

$$H^{(k)}(t) = \frac{1}{2} \sum_{a=0}^k \binom{k}{a} \operatorname{Re} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{k-a}(t) \right\rangle_{L^2}$$

and Cauchy–Schwarz in time give (270) and (275). More generally,

$$\partial_t^\ell E_{j+1/2} = \frac{1}{2} \sum_{a=0}^\ell \binom{\ell}{a} \operatorname{Re} \left\langle \Lambda^{j+1/2} V_a, \Lambda^{j+1/2} V_{\ell-a} \right\rangle_{L^2}.$$

Cauchy–Schwarz in time and the pointwise estimates (194)–(195) prove (273). Differentiating $\mathcal{P}_{j+1/2} = \partial_t E_{j+1/2} + 2\nu E_{j+3/2}$ then proves (274). The next two estimates follow from

$$\mathcal{P}_{1/2} = H' + \nu \left\| \Lambda^{3/2} u \right\|_2^2.$$

Finally,

$$|H(T) - H(t)| \leq \frac{1}{2} (\|u(T)\|_{\dot{H}^{1/2}} + \|u(t)\|_{\dot{H}^{1/2}}) \|u(T) - u(t)\|_{\dot{H}^{1/2}},$$

and the Banach identity in H^1 prove the square-root branch of (278). The identity

$$H' = \operatorname{Re} \left\langle \Lambda^{1/2} V_1, \Lambda^{1/2} V_0 \right\rangle_{L^2},$$

together with (195), proves its linear branch.

Sobolev embedding and (194) prove (279)–(281). Lemma 1.9 gives

$$\frac{d}{dt} \|u\|_{H^m}^2 \leq 2K_m \|\nabla u\|_\infty \|u\|_{H^m}^2.$$

Gronwall’s inequality and (280) prove (282). □

6 Three-dimensional applications

6.1 Explicit closure for small critical data

The construction below gives an explicit three-dimensional $\dot{H}^{1/2}$ realization of the critical-data mechanism introduced by Fujita and Kato [2]; every constant used here is derived in the displayed estimates.

Use the explicit homogeneous Sobolev constants

$$S_6 := S(6), \quad S_3 := S(3), \quad C_* := S_6 S_3.$$

Lemma 1.3 gives $C_* < \infty$ and prints its exact gamma-function value. The same constants hold for vector fields. Indeed, apply the scalar inequality to the Euclidean length $|F|$: one has $|\nabla|F|| \leq |\nabla F|$,

while the difference-quotient representation of $\dot{H}^{1/2}$ and $||F(x)| - |F(y)|| \leq |F(x) - F(y)|$ give $|||F|||_{\dot{H}^{1/2}} \leq \|F\|_{\dot{H}^{1/2}}$.

For every $z \in \mathcal{S}_\sigma(\mathbb{R}^3)$, define the critical nonlinear production by

$$\mathcal{P}_{1/2}(z) := -\operatorname{Re} \left\langle \Lambda^{1/2} z, \Lambda^{1/2} \mathbb{P} \nabla \cdot (z \otimes z) \right\rangle_{L^2}. \quad (283)$$

Lemma 6.1 (Critical transfer estimate). *Every $z \in \mathcal{S}_\sigma(\mathbb{R}^3)$ satisfies*

$$|\mathcal{P}_{1/2}(z)| \leq C_* \left\| \Lambda^{1/2} z \right\|_2 \left\| \Lambda^{3/2} z \right\|_2^2. \quad (284)$$

The same estimate holds for every divergence-free $z \in \dot{H}^{1/2}(\mathbb{R}^3) \cap \dot{H}^{3/2}(\mathbb{R}^3)$.

Proof. Since Λz is divergence free, the pressure term has zero pairing with Λz . Thus

$$\mathcal{P}_{1/2}(z) = -\operatorname{Re} \langle \Lambda z, (z \cdot \nabla) z \rangle_{L^2}.$$

Hölder's inequality and the definitions of $\mathbf{S}_6, \mathbf{S}_3$ give

$$|\mathcal{P}_{1/2}(z)| \leq \|z\|_6 \|\nabla z\|_2 \|\Lambda z\|_3 \leq C_* \|\Lambda z\|_2^2 \left\| \Lambda^{3/2} z \right\|_2.$$

Fourier-space Cauchy–Schwarz gives

$$\|\Lambda z\|_2^2 \leq \left\| \Lambda^{1/2} z \right\|_2 \left\| \Lambda^{3/2} z \right\|_2,$$

which proves (284). Solenoidal Fourier truncation followed by mollification approximates every field in the stated homogeneous intersection simultaneously in $\dot{H}^{1/2}$ and $\dot{H}^{3/2}$. Each factor in the estimate converges in its displayed space, so passage to the limit proves the final assertion. $\square$

For $m \in \mathbb{N}$, define the explicit homogeneous commutator constant

$$\mathbf{K}_m^{\text{hom}} := 3^m \sum_{\ell=1}^m \binom{m}{\ell} \mathbf{S}(\mathbf{p}_{m,\ell}) \mathbf{S}(\mathbf{q}_{m,\ell}), \quad (285)$$

where $(\mathbf{p}_{1,1}, \mathbf{q}_{1,1}) = (3, 6)$, and, for $m \geq 2$,

$$\vartheta_{m,\ell} := \frac{\ell-1}{m-1}, \quad \frac{1}{\mathbf{p}_{m,\ell}} := \frac{1}{3} - \frac{\vartheta_{m,\ell}}{6}, \quad \frac{1}{\mathbf{q}_{m,\ell}} := \frac{1}{6} + \frac{\vartheta_{m,\ell}}{6}.$$

The corresponding estimate is

$$|\operatorname{Re} \langle \Lambda^m \mathbb{P} \nabla \cdot (z \otimes z), \Lambda^m z \rangle_{L^2}| \leq \mathbf{K}_m^{\text{hom}} \left\| \Lambda^{3/2} z \right\|_2 \|\Lambda^m z\|_2 \|\Lambda^{m+1} z\|_2 \quad (286)$$

for every $z \in \mathcal{S}_\sigma(\mathbb{R}^3)$.

Lemma 6.2 (Explicit homogeneous critical commutator). *For every $m \in \mathbb{N}$, $\mathbf{K}_m^{\text{hom}} < \infty$, and (286) holds with the displayed constant. Moreover, (286) extends to every divergence-free $z \in H^{m+1}(\mathbb{R}^3) \cap \dot{H}^{3/2}(\mathbb{R}^3)$.*

Proof. The multinomial identity gives the exact pairing

$$\langle \Lambda^m f, \Lambda^m g \rangle_{L^2} = \sum_{|\alpha|=m} \frac{m!}{\alpha!} \langle \partial^\alpha f, \partial^\alpha g \rangle_{L^2}.$$

After differentiating $(z \cdot \nabla)z$, the undifferentiated transport term has zero pairing by incompressibility. A remaining term has the form

$$\left\langle (\partial^\beta z \cdot \nabla) \partial^{\alpha-\beta} z, \partial^\alpha z \right\rangle_{L^2}, \quad |\alpha| = m, \quad 0 < \beta \leq \alpha.$$

Put $\ell = |\beta|$. For $m \geq 2$, Hölder's inequality, Lemma 1.3, and Fourier log-convexity give

$$\begin{aligned} & \left\| \partial^\beta z \right\|_{\mathfrak{p}_{m,\ell}} \left\| \nabla \partial^{\alpha-\beta} z \right\|_{\mathfrak{q}_{m,\ell}} \\ & \leq S(\mathfrak{p}_{m,\ell}) S(\mathfrak{q}_{m,\ell}) \|z\|_{\dot{H}^{3/2}} \|z\|_{\dot{H}^{m+1}}. \end{aligned}$$

Indeed, the two differentiated Sobolev orders are the complementary interpolants

$$\begin{aligned} \ell + s(\mathfrak{p}_{m,\ell}) &= (1 - \vartheta_{m,\ell}) \frac{3}{2} + \vartheta_{m,\ell}(m+1), \\ m - \ell + 1 + s(\mathfrak{q}_{m,\ell}) &= \vartheta_{m,\ell} \frac{3}{2} + (1 - \vartheta_{m,\ell})(m+1). \end{aligned}$$

For $m = 1$, the same bound is the direct L^3 - L^6 estimate. Finally,

$$\sum_{|\alpha|=m} \frac{m!}{\alpha!} = 3^m, \quad \sum_{\substack{\beta \leq \alpha \\ |\beta|=\ell}} \binom{\alpha}{\beta} = \binom{m}{\ell}.$$

Summation gives exactly (285) and proves (286). Solenoidal Fourier truncation and convergence in $H^{m+1} \cap \dot{H}^{3/2}$ give the asserted extension. $\square$

Theorem 6.3 (Explicit small-critical-data global regularity). *Let $\nu > 0$ and $a \in \mathcal{S}_\sigma(\mathbb{R}^3)$. Set*

$$A_0 := \left\| \Lambda^{1/2} a \right\|_2, \quad \gamma_0 := \nu - C_* A_0.$$

If

$$A_0 \leq \frac{\nu}{2C_*}, \tag{287}$$

then a generates a unique global pressure-normalized smooth solution

$$(u, p) \in C^\infty([0, \infty) \times \mathbb{R}^3; \mathbb{R}^3 \times \mathbb{R}).$$

Uniqueness holds among pressure-normalized solutions whose velocity belongs to $C([0, T]; H_\sigma^3(\mathbb{R}^3))$ on every finite interval. For every $t \geq 0$,

$$\left\| \Lambda^{1/2} u(t) \right\|_2^2 + 2\gamma_0 \int_0^t \left\| \Lambda^{3/2} u(s) \right\|_2^2 ds \leq A_0^2, \quad \gamma_0 \geq \frac{\nu}{2}. \tag{288}$$

The critical nonlinear source and the first temporal response satisfy

$$\int_0^\infty \left\| \mathbb{P} \nabla \cdot (u \otimes u)(t) \right\|_{\dot{H}^{-1/2}}^2 dt \leq \frac{A_0^4}{4\pi^2 \gamma_0}, \tag{289}$$

$$\nu^2 \int_0^\infty \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt + \int_0^\infty \|u_t(t)\|_{\dot{H}^{-1/2}}^2 dt \leq \nu A_0^2 + \frac{A_0^4}{4\pi^2 \gamma_0}. \quad (290)$$

For every $m \in \mathbb{N}$, define

$$\Gamma_m^{\text{sm}} := \frac{(\mathcal{K}_m^{\text{hom}})^2 A_0^2}{2\nu \gamma_0}.$$

Then

$$\begin{aligned} & \|\Lambda^m u(t)\|_2^2 + \nu \int_0^t \|\Lambda^{m+1} u(s)\|_2^2 ds \\ & \leq \|\Lambda^m a\|_2^2 \exp(\Gamma_m^{\text{sm}}) \leq \|\Lambda^m a\|_2^2 \exp\left(\frac{(\mathcal{K}_m^{\text{hom}})^2 A_0^2}{\nu^2}\right). \end{aligned} \quad (291)$$

In particular,

$$\sup_{t \geq 0} \|u(t)\|_{H^m}^2 \leq 2^{m-1} \left[\|a\|_2^2 + \|\Lambda^m a\|_2^2 \exp(\Gamma_m^{\text{sm}}) \right]. \quad (292)$$

Proof. Iterate Theorem 1.10 and join overlapping intervals by its full $C([0, T]; H_\sigma^3)$ uniqueness. The union is a unique maximal H^3 solution on $[0, T_*)$, and persistence retains H^∞ on every compact subinterval. Put

$$A(t) := \left\| \Lambda^{1/2} u(t) \right\|_2, \quad B(t) := \left\| \Lambda^{3/2} u(t) \right\|_2.$$

The exact height balance and Lemma 6.1 give

$$\frac{1}{2} \frac{d}{dt} A(t)^2 + (\nu - C_* A(t)) B(t)^2 \leq 0.$$

Continuity and the first-exit argument give $A(t) \leq A_0$ throughout $[0, T_*)$. Integration proves (288).

Dual Sobolev, Hölder, and the unit Sobolev multiplier norm of $\mathbb{P}$ give

$$\|\mathbb{P} \nabla \cdot (u \otimes u)\|_{\dot{H}^{-1/2}} \leq S_3^3 \left\| \Lambda^{1/2} u \right\|_2 \left\| \Lambda^{3/2} u \right\|_2.$$

Since $S_3^6 = 1/(2\pi^2)$, the bound $A(t) \leq A_0$ and (288) imply (289). Integrating the exact identity (67), discarding its nonnegative terminal stock, and using (289) prove (290).

At order m , Lemma 6.2 and Young's inequality yield

$$\frac{d}{dt} \|\Lambda^m u\|_2^2 + \nu \|\Lambda^{m+1} u\|_2^2 \leq \frac{(\mathcal{K}_m^{\text{hom}})^2}{\nu} B(t)^2 \|\Lambda^m u\|_2^2.$$

Multiplication by the corresponding integrating factor and

$$\int_0^t B(s)^2 ds \leq \frac{A_0^2}{2\gamma_0}$$

prove (291). The Fourier inequality

$$(1 + |\xi|^2)^m \leq 2^{m-1} (1 + |\xi|^{2m})$$

and the kinetic-energy identity prove (292).

Let M_3 be the square root of the right-hand side of (292) with $m = 3$, and set

$$\delta_{\text{crit}} := \min\{1, c_\nu(1 + M_3)^{-2}\} > 0.$$

Every datum $u(t_0)$, $t_0 < T_*$, has H^3 -norm at most M_3 . Theorem 1.10 restarts the solution for at least δ_{crit} units of time. If $T_* < \infty$, choose $t_0 < T_*$ with $T_* - t_0 < \delta_{\text{crit}}/2$. Strong uniqueness joins the restarted solution to the original history and extends it beyond T_* , contradicting maximality. Hence $T_* = \infty$, and persistence gives the asserted smoothness. The same local uniqueness on each finite interval proves the stated global uniqueness class. $\square$

For $m \geq 1$, define

$$U_m^{\text{sm}} := 2^{(m-1)/2} \left[\|a\|_2^2 + \|\Lambda^m a\|_2^2 \exp(\Gamma_m^{\text{sm}}) \right]^{1/2}, \quad U_0^{\text{sm}} := \|a\|_2, \quad U_{-1}^{\text{sm}} := U_0^{\text{sm}}.$$

For $n \in \mathbb{N}_0$ and $m \geq -1$, define the finite-order recursion

$$\Upsilon_{0,m}^{\text{sm}} := U_m^{\text{sm}}, \quad (293)$$

$$\Upsilon_{n+1,m}^{\text{sm}} := \nu \Upsilon_{n,m+2}^{\text{sm}} + A_{m+1} \sum_{a'=0}^n \binom{n}{a'} \Upsilon_{a',m+2}^{\text{sm}} \Upsilon_{n-a',m+2}^{\text{sm}}. \quad (294)$$

Proposition 6.4 (Explicit small-data pressure-complete rectangles). *Under the hypotheses of Theorem 6.3, for every $n \in \mathbb{N}_0$ and $m \geq -1$,*

$$\sup_{t \geq 0} \|V_n(t)\|_{H^m} \leq \Upsilon_{n,m}^{\text{sm}}. \quad (295)$$

For every $T < \infty$ and $N, M \in \mathbb{N}_0$,

$$\mathfrak{R}_{N,M}(u; T) \leq T \sum_{n=0}^N [(\Upsilon_{n,M+1}^{\text{sm}})^2 + (\Upsilon_{n+1,M-1}^{\text{sm}})^2], \quad (296)$$

$$\mathfrak{Q}_{N,M}(p; T) \leq T A_M \sum_{n=0}^N \sum_{a'=0}^n \binom{n}{a'} \Upsilon_{a',M+1}^{\text{sm}} \Upsilon_{n-a',M+1}^{\text{sm}}. \quad (297)$$

In the entrywise order from Proposition 5.9, these bounds have the matrix form

$$\mathbf{C}_{N,M}(T) \preceq_{\text{ent}} \begin{pmatrix} T^{1/2} \Upsilon_{n,M+1}^{\text{sm}} & T^{1/2} \Upsilon_{n+1,M-1}^{\text{sm}} & T A_M \sum_{a'=0}^n \binom{n}{a'} \Upsilon_{a',M+1}^{\text{sm}} \Upsilon_{n-a',M+1}^{\text{sm}} \end{pmatrix}_{n=0}^N. \quad (298)$$

Every right-hand side is an explicit finite function of ν, T, N, M and the finite set of datum norms appearing in U_m^{sm} along the recurrence (294).

Proof. Equation (292) proves the base (295) for $m \geq 1$; the kinetic-energy identity proves it for $m = 0$, and $H^0 \hookrightarrow H^{-1}$ proves it for $m = -1$. Apply $\mathbb{P}$ to the temporal recurrence. This is the $\alpha = 0$ temporal block-row of (30); its factorial conjugate is (33). Equation (172) at order $m+1$ and induction give (295) through (294); for $m = -1$, use the order-zero product estimate and $H^0 \hookrightarrow H^{-1}$. Integration in time gives (296). The pressure recurrence and (173) at order M , followed by integration and summation in n , give (297). The same rowwise bounds give (298); taking the Frobenius norm of its first two columns and summing its pressure column recovers (296) and (297). $\square$

6.2 Relative energy and the Leray–Hopf class

The weak class is stated in the whole-space form originating with Leray and Hopf [4, 3]. The argument below derives its stability against the pressure-complete smooth history directly from the weak formulation and the energy inequality.

Definition 6.5 (Three-dimensional Leray–Hopf solution). Let $\nu > 0$, $T > 0$, and $v_0 \in H_\sigma^0(\mathbb{R}^3)$. A Leray–Hopf solution on $[0, T]$ is a pair (v, q) such that

$$v \in C_w([0, T]; H_\sigma^0) \cap L^\infty(0, T; L_\sigma^2) \cap L^2(0, T; \dot{H}_\sigma^1), \quad q \in \mathcal{D}'((0, T) \times \mathbb{R}^3),$$

the equations

$$\partial_t v + (v \cdot \nabla) v + \nabla q = \nu \Delta v, \quad \nabla \cdot v = 0$$

hold in distributions, $v(t) \rightarrow v_0$ strongly in L^2 as $t \downarrow 0$, and

$$\frac{1}{2} \|v(t)\|_2^2 + \nu \int_0^t \|\nabla v(s)\|_2^2 \, ds \leq \frac{1}{2} \|v_0\|_2^2 \quad (299)$$

for every $t \in [0, T]$. Every divergence-free test field $\phi \in C_c^\infty([0, T] \times \mathbb{R}^3; \mathbb{R}^3)$ satisfies

$$\int_0^T \left[-\langle v, \partial_t \phi \rangle - \int_{\mathbb{R}^3} (v \otimes v) : \nabla \phi \, dx + \nu \langle \nabla v, \nabla \phi \rangle \right] dt = \langle v_0, \phi(0) \rangle. \quad (300)$$

A global Leray–Hopf solution at viscosity ν is a pair on $[0, \infty)$ whose restriction to every finite $[0, T]$ is a Leray–Hopf solution in Definition 6.5, with the same datum and viscosity.

Lemma 6.6 (Solenoidal cutoff density and time commutation). *Let $\mathcal{A}$ be a finite subset of $\mathbb{N}_0 \times (1, \infty)$. There are linear operators $\mathcal{T}_k$, independent of time, such that, for every $f \in H_\sigma^\infty(\mathbb{R}^3)$,*

$$\mathcal{T}_k f \in C_{c,\sigma}^\infty(\mathbb{R}^3)$$

and, for every $(r, p) \in \mathcal{A}$,

$$\sup_k \|\mathcal{T}_k\|_{\mathcal{L}(W^{r,p}, W^{r,p})} < \infty, \quad (301)$$

$$\|\mathcal{T}_k f - f\|_{W^{r,p}} \longrightarrow 0. \quad (302)$$

The convergence is uniform for f in a compact subset of the retained intersection of $W^{r,p}$ spaces. If $F \in C^\infty([0, T]; H_\sigma^\infty)$, then

$$\partial_t^a (\mathcal{T}_k F) = \mathcal{T}_k (\partial_t^a F) \quad (a \in \mathbb{N}_0),$$

and the convergence in (302) is uniform in t for every fixed temporal derivative.

Proof. Choose $\chi \in C_c^\infty(B_2)$ with $\chi = 1$ on B_1 , and put $\chi_R(x) = \chi(x/R)$ and $A_R = B_{2R} \setminus \overline{B_R}$. For a smooth solenoidal f , set

$$g_R := \nabla \chi_R \cdot f.$$

It is supported in A_R , and

$$\int_{A_R} g_R \, dx = \int_{\mathbb{R}^3} \nabla \cdot (\chi_R f) \, dx = 0.$$

Cover the fixed annulus A_1 by finitely many star-shaped coordinate patches, choose a subordinate partition of unity, and apply the explicit Bogovskii integral operator on each patch. After the standard successive mean correction on the overlaps, this gives a linear operator $\mathcal{G}_1 : C_{c,0}^\infty(A_1) \rightarrow C_c^\infty(A_1; \mathbb{R}^3)$ satisfying

$$\nabla \cdot \mathcal{G}_1 g = g, \quad \|\nabla^{a+1} \mathcal{G}_1 g\|_{L^p(A_1)} \leq C_{a,p} \|\nabla^a g\|_{L^p(A_1)} \quad (a \in \mathbb{N}_0).$$

Scaling $x = Ry$ gives $\mathcal{G}_R$ on A_R with

$$\begin{aligned} \|\mathcal{G}_R g\|_{L^p(A_R)} &\leq C_p R \|g\|_{L^p(A_R)}, \\ \|\nabla^{a+1} \mathcal{G}_R g\|_{L^p(A_R)} &\leq C_{a,p} \|\nabla^a g\|_{L^p(A_R)}. \end{aligned} \quad (303)$$

Define

$$\mathcal{S}_R f := \chi_R f - \mathcal{G}_R g_R.$$

Then $\mathcal{S}_R f \in C_{c,\sigma}^\infty(B_{2R})$. Leibniz' rule gives, for $0 \leq a \leq r-1$,

$$\|\nabla^a g_R\|_{L^p(A_R)} \leq C_a \sum_{b=0}^a R^{-1-a+b} \|\nabla^b f\|_{L^p(A_R)}.$$

Combining this estimate with (303) and the analogous cutoff estimate yields

$$\|\mathcal{S}_R f - f\|_{W^{r,p}} \leq C_{r,p} \sum_{b=0}^r (1 + R^{b-r}) \|\nabla^b f\|_{L^p(\mathbb{R}^3 \setminus B_R)}. \quad (304)$$

For $r = 0$, the same conclusion follows from the first line of (303) and $\|g_R\|_p \leq CR^{-1} \|f\|_{L^p(A_R)}$. The right side tends to zero as $R \rightarrow \infty$, and the same calculations give operator norms bounded independently of R .

Choose $R_k \uparrow \infty$ and set $\mathcal{T}_k := \mathcal{S}_{R_k}$. Estimate (304) holds simultaneously for the finite family $\mathcal{A}$, while the preceding operator estimates give (301) simultaneously for that family. Strong convergence plus the uniform operator bound makes convergence uniform on compact subsets: cover a compact set by finitely many norm balls and use the bound on each remainder. Each $\mathcal{T}_k$ is independent of time, so differentiation commutes with it. The ranges of $\partial_t^a F$ on $[0, T]$ are compact in every retained space, which proves the final assertion. $\square$

Lemma 6.7 (Admissible strong-history test). *Let u be the global solution from Theorem 4.8, and let v be a Leray–Hopf solution on $[0, T]$ at the same viscosity. Put $w := v - u$. Then, for every $t \in [0, T]$,*

$$\begin{aligned} \langle v(t), u(t) \rangle &= \langle v_0, u_0 \rangle + \int_0^t \int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot w \, dx \, ds \\ &\quad - 2\nu \int_0^t \langle \nabla v, \nabla u \rangle \, ds. \end{aligned} \quad (305)$$

Proof. The energy class and spatial interpolation give

$$v \in L^4(0, T; L^3), \quad \|v\|_{L^4 L^3}^4 \leq C \|v\|_{L^\infty L^2}^2 \|\nabla v\|_{L^2 L^2}^2,$$

so $v \otimes v \in L^2(0, T; L^{3/2})$.

Fix $0 < t < T$. Solenoidal density on $\mathbb{R}^3$, applied uniformly to the compact sets $u([0, T])$ and $u_t([0, T])$ in their finite Sobolev spaces, gives

$$u^{(k)} \in C^\infty([0, T]; C_{c,\sigma}^\infty(\mathbb{R}^3))$$

such that

$$u^{(k)} \longrightarrow u \quad \text{in} \quad W^{1,1}(0, T; L^2) \cap L^2(0, T; H^1) \cap L^2(0, T; W^{1,3}) \cap C([0, T]; L^2). \quad (306)$$

Apply Lemma 6.6 with

$$\mathcal{A} = \{(0, 2), (1, 2), (1, 3)\}$$

and set $u^{(k)} = \mathcal{T}_k u$. Its uniform compact-range conclusion for u and u_t , together with time commutation (6.6), gives (306).

For $0 < h < T - t$, choose $\eta_h \in C_c^\infty([0, T])$ with $0 \leq \eta_h \leq 1$, $\eta_h = 1$ on $[0, t]$, $\eta_h = 0$ on $[t + h, T]$, and $\eta_h' \leq 0$. The field $\eta_h u^{(k)}$ is an admissible test in (300). First let $k \rightarrow \infty$. The convergence in (306), together with $v \otimes v \in L^2 L^{3/2}$, permits passage in every term. Next choose the cutoffs so that

$-\eta'_h$ is an approximate identity from the right at t , and let $h \downarrow 0$. Weak continuity of v , strong continuity of u , and the Lebesgue differentiation theorem give

$$\begin{aligned} \langle v(t), u(t) \rangle &= \langle v_0, u_0 \rangle + \int_0^t \langle v, u_t \rangle \, ds + \int_0^t \int_{\mathbb{R}^3} (v \otimes v) : \nabla u \, dx \, ds \\ &\quad - \nu \int_0^t \langle \nabla v, \nabla u \rangle \, ds. \end{aligned}$$

This identity first holds for almost every t . Its right-hand side is continuous in t , while $t \mapsto \langle v(t), u(t) \rangle$ is continuous by weak continuity of v and norm continuity of u . Hence it holds for every $0 \leq t \leq T$, including $t = T$ by passage from below. Using

$$u_t = -(u \cdot \nabla)u - \nabla p + \nu \Delta u$$

is admissible because v is divergence free. It gives

$$\int_0^t \langle v, u_t \rangle \, ds = - \int_0^t \int_{\mathbb{R}^3} v \cdot (u \cdot \nabla)u \, dx \, ds - \nu \int_0^t \langle \nabla v, \nabla u \rangle \, ds.$$

Finally,

$$\int_{\mathbb{R}^3} [(v \otimes v) : \nabla u - v \cdot (u \cdot \nabla)u] \, dx = \int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot w \, dx,$$

because

$$\int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot u \, dx = \frac{1}{2} \int_{\mathbb{R}^3} w \cdot \nabla |u|^2 \, dx = 0.$$

This proves (305). $\square$

Theorem 6.8 (Quantitative Leray–Hopf stability). *Let $\nu > 0$ and $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, and let u be the global strong history from Theorem 4.8. Fix $T > 0$, and let v be any Leray–Hopf solution on $[0, T]$ at the same viscosity ν , with datum $v_0 \in H_\sigma^0(\mathbb{R}^3)$. Then, for $0 \leq t \leq T$,*

$$\begin{aligned} \frac{1}{2} \|v(t) - u(t)\|_2^2 + \nu \int_0^t \|\nabla(v - u)(s)\|_2^2 \, ds \\ \leq \frac{1}{2} \|v_0 - u_0\|_2^2 \exp\left(2 \int_0^t \|\nabla u(s)\|_\infty \, ds\right) \end{aligned} \tag{307}$$

$$\leq \frac{1}{2} \|v_0 - u_0\|_2^2 \exp\left(2S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}[u_0, \nu; T]^{1/2}\right). \tag{308}$$

Proof. Combine (299), the strong energy equality (167), and (305). With $w = v - u$, this gives

$$\frac{1}{2} \|w(t)\|_2^2 + \nu \int_0^t \|\nabla w(s)\|_2^2 \, ds \leq \frac{1}{2} \|w(0)\|_2^2 - \int_0^t \int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot w \, dx \, ds. \tag{309}$$

The last integrand has absolute value at most

$$\|\nabla u(s)\|_\infty \|w(s)\|_2^2.$$

Gronwall's inequality bounds $\|w(t)\|_2^2$. Substitution of that bound back into (309) proves (307). Proposition 5.12 gives

$$\int_0^T \|\nabla u(t)\|_\infty \, dt \leq S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}[u_0, \nu; T]^{1/2},$$

which proves (308). $\square$

Corollary 6.9 (Explicit weak-class stability for small critical data). *Under the hypotheses of Theorem 6.3, let v be a global Leray–Hopf solution at the same viscosity ν , with datum $v_0 \in H_\sigma^0(\mathbb{R}^3)$. Then*

$$\begin{aligned} \frac{1}{2} \|v(t) - u(t)\|_2^2 + \nu \int_0^t \|\nabla(v - u)(s)\|_2^2 \, ds \\ \leq \frac{1}{2} \|v_0 - a\|_2^2 \exp(2S_{3,1,\infty} t U_3^{\text{sm}}) \quad (t \geq 0). \end{aligned} \quad (310)$$

Proof. Equation (292) gives $\sup_{s \geq 0} \|u(s)\|_{H^3} \leq U_3^{\text{sm}}$. Hence

$$\int_0^t \|\nabla u(s)\|_\infty \, ds \leq S_{3,1,\infty} t U_3^{\text{sm}}.$$

Apply (307). □

Corollary 6.10 (Uniqueness and energy equality in the Leray–Hopf class). *Let $\nu > 0$ and $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$. Every three-dimensional Leray–Hopf solution that is global at viscosity ν , with datum u_0 , agrees on every finite interval with the global solution from Theorem 4.8. Its velocity has the smooth representative*

$$u \in C^\infty([0, \infty) \times \mathbb{R}^3; \mathbb{R}^3),$$

and it satisfies

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \, ds = \frac{1}{2} \|u_0\|_2^2 \quad (t \geq 0).$$

Any pressure representative normalized by decay at spatial infinity equals the pressure in (166).

Proof. Set $v_0 = u_0$ in Theorem 6.8. The right-hand side of (308) is zero, so $v = u$ in $C_w([0, T]; L^2) \cap L^2(0, T; \dot{H}^1)$. Since T is arbitrary, the identity holds globally. Smoothness and energy equality follow from (163) and (167). The pressure claim follows by taking the divergence of the momentum equation and applying the normalization. □

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